

TR DD MC System Path Troubleshooting

[\(Click Here for Summary Version\)](#)

1 Personnel Requirements

Required Persons	Preliminary Reqs	Procedure	Finalization
1	Not Applicable	60 mins	20 mins

2 Overview

System configurations can be different depending on whether the system is a forward production HDx system, or an upgrade.

- For all references to **“Upgrade Systems”**, this refers to systems with the old LPCA, including the green Bendix Legacy Connector and a Head T/R Switch. These systems have two Head T/R Lines, one connecting to the Head T/R Switch, and the other for Connector **A**.
- Forward Production systems** have the new LPCA, which no longer includes the Head T/R switch, as this circuitry has been moved into the coils themselves. The new LPCA also has a Hypertronics A connector in place of the Bendix Legacy connector, and this interfaces with the Multi-coil Interface Box.
- For a picture of the different LPCAs, please refer to the [Multi-coil Receive Chain \(MCR\) Tools](#)

(For Troubleshooting Process)

- Once an error is found, Refer to [Table 1](#), for the fault in question. For those problems with links in the T/S Action column, select the link, to be taken to the troubleshooting section in this document corresponding to that fault.

NOTE: If the complete troubleshooting process is followed, but the problem is not found because it is no longer occurring, the problem is most likely an intermittent cable connection issue. The next time the fault occurs, it may be warranted to run the diagnostic after every step with the system in the normal condition, to determine what cable connection may have the intermittent issue.

Table 1: TR DD MC Problem Description and Troubleshooting Action

Fault	Problem Description	Field Action
T/R Open Circuit on: - Body Path - Head Path - Conn A Head Path - MNS Path	TR Open Circuit Fault – system error log will give additional information on Source of fault, measured Value, and Limit in mA.	TR Open Circuit Fault – Body TR Open Circuit Fault – Head TR Open Circuit Fault – MNS
T/R Short Circuit on: - Body Path - Head Path - Conn A Head Path - MNS Path	TR Short Circuit Fault – system error log will give additional information on Source of fault, measured Value, and Limit in mA.	TR Short Circuit Fault – Body TR Short Circuit Fault – Head TR Short Circuit Fault – MNS
DD High Voltage (HV): - BD1 HV not w/in bounds - BD2 HV not w/in bounds	DD High Voltage Fault- system error log will give the Source of the Fault as	Dynamic Disable and Direct Drive High Voltage Fault

Fault	Problem Description	Field Action
- BD3 HV not w/in bounds - BD4 HV not w/in bounds - DD HV not w/in bounds	well as the measured Value and Limit in Volts.	
DD Open: - Limit on BD1 exceeded - Limit on BD2 exceeded - Limit on BD3 exceeded - Limit on BD4 exceeded - Limit on DD exceeded	DD Open Circuit Fault– system error log will give additional information on Source of fault, measured Value, and Limit in mV.	DD Open Circuit Fault
DD Short: - Limit on BD1 exceeded - Limit on BD2 exceeded - Limit on BD3 exceeded - Limit on BD4 exceeded - Limit on DD exceeded	DD Short Circuit Fault – system error log will give additional information on Source of fault, measured Value, and Limit in mA.	DD Short Circuit Fault
MC Open: Connector <X> - - MC Check fail. Short circuit on channels: <channels that failed>	MC Open Circuit Fault- Tool will give connector that failed (A,B,C, or L (legacy), as well as the channels that faulted on the connector.	MC Open Circuit Fault
MC Short: Connector <X> - MC Check fail. Short circuit on channels: <channels that failed>	MC Short Circuit Fault- Tool will give connector that failed (A,B,C or L (legacy), as well as the channels that faulted on the connector.	MC Short Circuit Fault
MC Receive: Connector <X> - MC Check fail. Short circuit on channels: <channels that failed>	MC Receive Fault- considered a short circuit. Tool will give connector that failed (A,B,C or L (legacy), as well as the channels that faulted on the connector.	MC Receive Fault
MC Transmit: Connector <X> - MC Check fail. Open circuit on channels: <channels that failed>	MC Transmit Fault- considered an open circuit. Tool will give connector that failed (A,B,C or L (legacy), as well as the channels that faulted on the connector.	MC Transmit Fault
Head and Body Cables Possibly Swapped	System is detecting current on the Head and Body T/R cables that is consistent with them being swapped.	Head and Body TR Cable Swap: Swap head TR and body TR cables at Driver Module or input to the RF Amp.
Head and Connector A T/R Cables Possibly Swapped	System is detecting current on the Head and Connector A cables that is consistent with them being swapped. This error would only occur in an upgrade system, which still has the old LPCA with a Legacy connector	Head and Connector A Cable Swap: Swap head and Conn A TR cables at the TR switch on the pen wall, or Cable I/F at Rear Ped.
MC Cables Swapped	System is detecting that the MC Cables are potentially swapped.	MC Cable Swap: Swap the MC Cables at the Driver Module, Pen-panel, or 16-channel switch.
MC Cables Disconnected	The Driver Module has detected that Multicoil Driver Cable <1,2> is disconnected.	MC Cable Disconnected: Check for Disconnected cables at the Driver Module, Pen-panel, or 16-channel switch.

3 Preliminary Requirements

3.1 Tools and Test Equipment

Item	Qty	Effectivity	Part#	Manufacturer
50 Ohm Dummy Load	1	-	2372868	-
Small Philips head screwdriver (size #0 or #00) with at least 2 inch length	1	-	-	-
Digital Multi Meter	1	-	-	-
RF Test Cables Kit	1	-	-	-
MCR Tool (Type and Part # dependent on system type)	As Directed in Troubleshooting Sections	-	-	-

3.2 Safety



DANGER

FATAL ELECTRIC SHOCK HAZARD!!

FATAL ELECTRICAL VOLTAGES THROUGH OUT THE RF AND MNS CABINETS.

TO PREVENT FATAL ELECTRIC SHOCK, MAKE SURE THE RF AMPLIFIERS ARE DISABLED WHEN CONFIGURING THE TROUBLESHOOTING TOOLS.



DANGER

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FATAL VOLTAGES FROM THE RF AND MNS AMPLIFIERS WHEN ENTERGIZED.

TO PREVENT FATAL ELECTRIC SHOCK, WHEN DIRECTED, TURN OF THE RFI/AMP AND MNS BREAKERS ON THE FRONT OF THE PDU BEFORE YOU PERFORM THE PROCEDURES. LOCK AND TAG OUT THE PLEXI-GLASS COVER ON THE FRONT OF THE PDU.



CAUTION

Strong Magnetic Field!

Ferrous materials can become dangerous projectiles in the presence of the magnetic field Produced by the Signa Magnet.

The Digital Multi Meter (DMM) has ferrous components. Use extreme caution if this tool must be brought into the magnet room, keeping it as far from the magnet as possible.

3.3 Required Conditions

Condition	Reference	Effectivity
Ensure no scan is running and the system is in the idle condition.	-	-
If the system is configured for multi-nuclear spectroscopy, please plug in a MNS TR (transmit-receive) switch if the fault is MNS related.	-	-

4 Procedure

4.1 TR Open Circuit Fault – Body

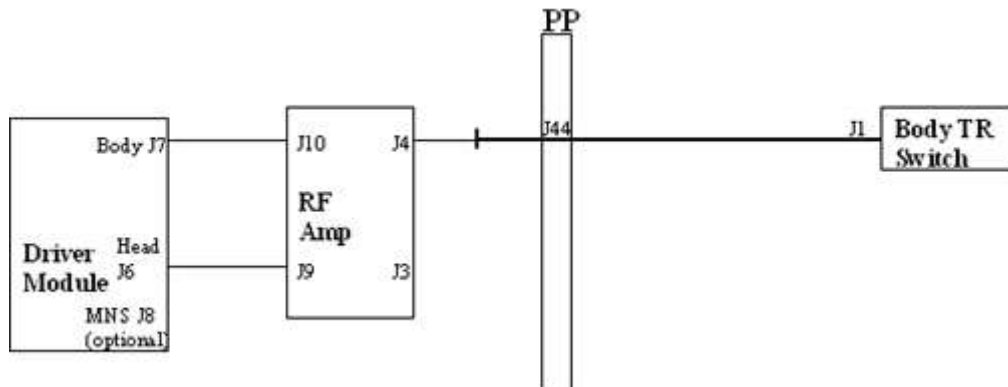
If the system is reporting a T/R Open Circuit Fault along the Body line, troubleshooting must occur to isolate it to the failed component. [Illustration 1](#) shows a path for the TR signals.

NOTE: In the unlikely situation both a Body and Head TR Open occur at the same time, verify the T/R cables have been connected throughout the chain. If Failure still occurs, check the power supplies in the Driver Module.

1. Swap the Head (J6) and Body (J7) T/R output cables at the Driver Module, and rerun the Diagnostic to determine if the failure changes.

NOTE: You will continue to have a Body T/R open fault in this scenario due to the difference in Head to Body output signals. The important factor is what happens with the Head T/R line.

- a. If the Body still reports a T/R Open, and the head does not report any failure, the problem is within the Driver Module. Replace the driver module, refer to the [Driver Module Replacement](#) procedure.
- b. If the Head now reports a T/R Open, the problem is somewhere further along the Body T/R path. Do **NOT** replace the Driver Module – continue to troubleshoot.
- c. Return the Driver Module Head and Body T/R output cables to their proper connection points.

Illustration 1: System Body T/R signal Path

2. Swap the Head (J9) and Body (J10) T/R Input cables at the RF Amplifier, and rerun the Diagnostic to determine if the failure changes.

NOTE: You will continue to have a Body T/R open in this scenario due to the difference in Head to Body output signals. The important factor is what happens with the Head T/R line.

- a. If the Body still reports a T/R Open, and the head does not report any failure, the problem is within the cable or connection between the Driver Module and RF Amplifier. Verify the connection integrity, and if all connections are good, but still getting an error, replace the cable.
- b. If the Head now reports a T/R Open, the problem is somewhere further along the Body T/R path. Continue to troubleshoot the other components.
- c. Return the Head and Body T/R RF Amplifier Input cables to their proper connection points.



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3. Obtain a 50ohm terminator and use a DMM to measure and record its resistance from center pin to shell.



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4. On the front of the PDU, Lock out and Tag out the RFI/AMP Breaker, to remove power from the RF Amplifier.
5. Disconnect the Body T/R cable at the Driver Module (J7), and connect the 50-ohm terminator to the cable.
6. Disconnect the Body Transmit cable at the Body TR Switch/Hybrid (J1), and use a DMM to measure the resistance between the center pin and the shell of the cable.
 - a. If the DMM measures within 10 ohms of the value recorded in [Step 3](#), the problem is with the Body Hybrid. Replace the Body Hybrid according to the following replacement procedures.

- b. If the DMM measures more than 10 ohms over the value recorded in [Step 3](#) then the problem is between the Body Hybrid Splitter and RF Amp.
 - c. Reconnect the cable to the Body Hybrid.
- 7. In the Equipment Room, locate the connection between the smaller Body Flexible RF Extension cable and the larger Body RF Transmit cable. This connection is often located under the Equipment Room floor.
- 8. Disconnect the small Body Flexible RF Extension cable from the larger Body RF Transmit cable, and using a DMM, measure the resistance between the center pin and the connector shell of the smaller Body Flex cable.
 - a. If the DMM measures within 10 ohms of the value recorded in [Step 3](#), the problem is somewhere on the larger Body RF Transmit Circuit. Continue troubleshooting this circuit at [Step 10](#).
 - b. If the DMM measures more than 10 ohms over the value recorded in [Step 3](#) then the problem is between the smaller Body Flexible Extension cable and RF Amp.
 - c. Reconnect the Body Flexible RF Extension cable to the larger Body RF Transmit cable.
- 9. Disconnect the Body Flexible RF Extension cable from J4 on the RF Amp, and use a DMM to measure the resistance between the center pin and outer shell of the J4 connector on the RF Amplifier.
 - a. If the DMM measures within 10 ohms of the value recorded in [Step 3](#), check the connectors on in the Body Flexible RF Extension cable. Repair or replace this cable.
 - b. If the DMM measures more than 10 ohms over the value recorded in [Step 3](#) then the problem appears to be the RF Amp.
 - c. Transfer the 50-ohm terminator from the J7 cable to J10 on the RF Amp, and reconnect the TR cable to J7 on the Driver Module.
 - d. Use a DMM and measure the resistance between the center pin and shell of J4 on the RF Amp.
 - i. If the DMM measures within 10 ohms of the value recorded in [Step 3](#) then recheck the condition of the connector on the J10 cable. Repair connector or replace cable.
 - ii. If the DMM measures more than 10 ohms over the value recorded in [Step 3](#) then troubleshoot the RF Amp, and replace if necessary.
- 10. Inspect the larger Body Transmit circuit.
 - a. Inspect RF connectors on each end of the cable. Make sure connectors are firmly attached to cable and center pins are not loose.
 - b. Remove cables from connectors at Penetration Panel.
 - c. Measure for low resistance between shields on each end of cables.
 - d. Measure for low resistance between center pins on each end of cables.
 - e. Inspect Pen Panel connection. Pay special attention to center pin connection.
 - f. If necessary re-terminate or repair cable.
- 11. Return all cables to their proper positions.

4.2 TR Open Circuit Fault – Head

If the system is reporting a T/R Open Circuit Fault along the Head path, troubleshooting must occur to isolate it to the failed component. [Illustration 2](#) shows a path for the TR signals. There are differences between an upgraded system, which still has the Legacy connector, and a forward production system with the new LPCA. An upgraded system utilizes a RF TX switch on the inside of the Pen-Panel to split the head signal into two. One supplies the Head Port, while the other supplies Connector **A**, as shown in [Illustration 2](#). In a forward production system, the splitter is bypassed, and not used, so there is only one head signal in the Magnet room. See [Illustration 3](#). Also the Head TR Switching circuitry is built into the Split Head coil cable connector.

1. Swap the Head (J6) and Body (J7) T/R output cables at the Driver Module, and rerun the Diagnostic to determine if the failure changes.

NOTE: You will also receive a Head and Body TR Cable Swap error when swapping the cables, if the true open is further down the Head T/R line.

- a. If the Head still reports a T/R Open, the problem is within the Driver Module. Replace the driver module, refer to the [Driver Module Replacement](#) procedure.
- b. If the Body now reports a T/R Open, and the Head does not report any failure, the problem is somewhere further along the Head T/R path. Do **NOT** replace the Driver Module – continue to troubleshoot.
- c. Return the Driver Module Head and Body T/R output cables to their proper connection points.

Illustration 2: Upgrade System Head T/R signal Path

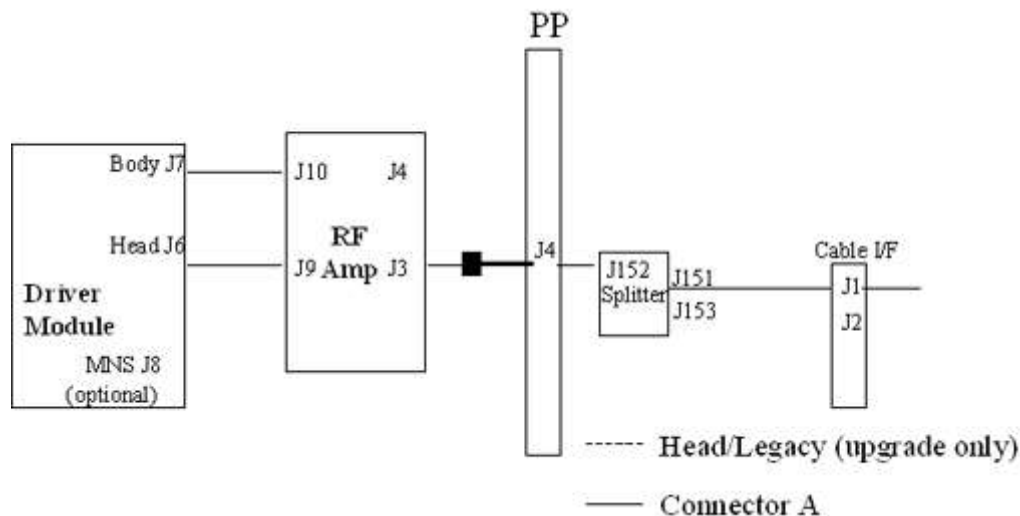
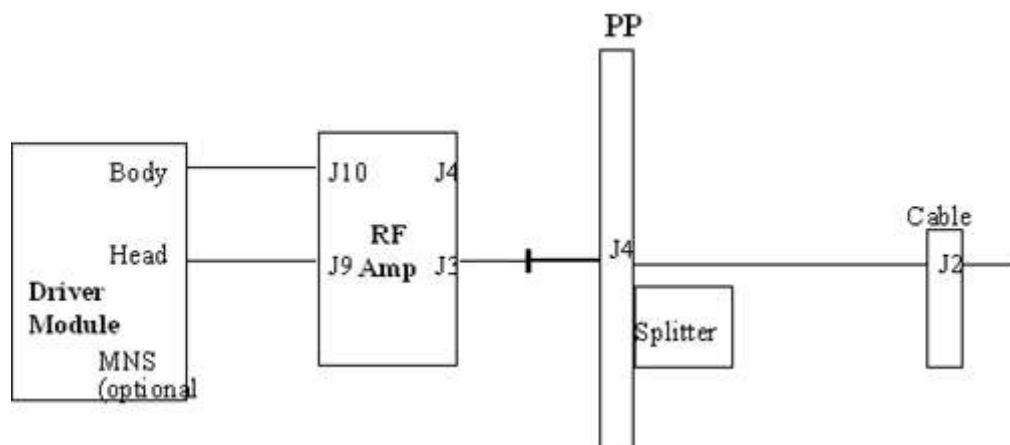


Illustration 3: Forward Production Head T/R signal Path



2. Swap the Head (J9) and Body (J10) T/R Input cables at the RF Amplifier, and rerun the Diagnostic to determine if the failure changes.

NOTE: You will also receive a Head and Body TR Cable Swap error when swapping the cables, if the true open is further down the Head T/R line.

- a. If the Head still reports a T/R Open, the problem is within the cable or connection between the Driver Module and RF Amplifier. Verify the connection integrity. Trace connection back to Driver Module and confirm that this really is Head TR cable and not Body TR cable. If all connections are good, but still getting an error, replace the cable.
- b. If the Body now reports a T/R Open, the problem is somewhere further along the Head T/R path. Continue to troubleshoot the other components.

- c. Return the Head and Body T/R RF Amplifier input cables to their proper connection points.



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- 3. Obtain a 50ohm terminator and use a DMM to measure and record its resistance from center pin to shell.



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- 4. On the front of the PDU, Lock out and Tag out the RFI/AMP Breaker, to remove power from the RF Amplifier.
- 5. Disconnect the Head T/R cable at the Driver Module (J6) and connect the 50-ohm terminator to the cable.
- 6. Disconnect the Head T/R output cable at the RF Amplifier (J3) and use a DMM to measure the resistance between the center conductor and shell of the J3 connector on the RF Amp.
 - a. If the DMM measures within 10 ohms of the value recorded in [Step 3](#), the problem is with one of the components in the RF transmit chain after the RF Amplifier. Continue with [Step 7](#).
 - b. If the DMM measures more than 10 ohms of the value recorded in [Step 3](#), the problem appears to be with the RF Amplifier. Re-inspect the cable connecting to J9 on the RF Amp and, if OK, repair or replace RF Amp.
 - c. Reconnect the cable to the RF Amplifier J3.
- 7. In the Equipment Room, locate the connection between the Head Flexible RF Extension cable and the Head RF Transmit cable. This connection is often located under the Equipment Room floor.
- 8. Disconnect the Head Flexible RF Extension cable from the Head RF Transmit cable, and using a DMM, measure the resistance between the center pin and the connector shell of the smaller Head Flexible RF Extension cable.
 - a. If the DMM measures within 10 ohms of the value recorded in [Step 3](#), the problem is somewhere further down the Transmit Chain.
 - b. If the DMM measures more than 10 ohms over the value recorded in [Step 3](#) then the problem appears to be in the Head Flexible RF Extension cable. Repair or replace the cable.
 - c. Reconnect the Head Flexible RF Extension cable to the Head RF Transmit cable.
- 9. Disconnect the Head Transmit cable at J4 on the Equipment Room side of the Penetration Panel.
- 10. Use a DMM to measure the resistance between the center pin and shield of the cable removed from J4.
 - a. If the DMM measures within 10 ohms of the value recorded in [Step 3](#) check the center pin on the connector. Make sure it is long enough to contact the center conductor in the Penetration Panel J4 feed-through connector. The problem appears to be with one of the components further down the transmit chain.
 - b. If the DMM measures more than 10 ohms over the value measure in [Step 3](#) then the problem is with the Head RF Transmit cable. Inspect the connectors on each end of the cable and re-terminate cable ends if necessary.
 - c. Reconnect cable to J4.

- 11. Disconnect the Head Transmit cable at J4 on the Magnet Room side of the Penetration Panel.

- 12. Use a DMM to measure the resistance between the center pin and shield of the Penetration Panel J4 feed-through connector.

- a. If the DMM measures within 10 ohms of the value recorded in [Step 3](#) the problem is with one of the components further down the transmit chain.
- b. If the DMM measures more than 10 ohms over the value measure in [Step 3](#) then replace Penetration Panel J4 feed-through connector.
- c. Reconnect cable to J4.

If this is a **forward production system** then skip to [Step 16](#).

13. **(For Upgrade system:)** Disconnect the Head Transmit cable at J152 on RF Transmit Switch and use a DMM to measure the resistance between the center pin and shield of the cable.
 - a. If the DMM measures within 10 ohms of the value recorded in [Step 3](#) check the center pin on the connector. Make sure it is long enough to contact the center conductor in J152 on RF Transmit Switch. The problem may be with one of the components further down the transmit chain.
 - b. If the DMM measures more than 10 ohms over the value measure in [Step 3](#) then the problem is with the cable. Make certain this cable comes directly from J4 on the Penetration Panel. If cable identity is OK then replace cable.
 - c. Reconnect cable to J152.

If this problem is occurring when a transmit/receive coil is attached to the green Head Bendix connector then skip to [Step 22](#). If coil attached to Connector **A** then proceed to [Step 14](#).
14. Connect failing transmit/receive coil to Connector A. This must be done or RF Transmit Switch will not toggle to correct port.
15. **(For Upgrade system:)** Disconnect the Conn A Head Transmit cable at J153 on RF Transmit Switch and use a DMM to measure the resistance between the center conductor and shield of J153 on the RF Transmit Switch.
 - a. If the DMM measures within 10 ohms of the value recorded in [Step 3](#) check the center pin on the cable connector. Make sure it is long enough to contact the center conductor in J153 on RF Transmit Switch. The problem may be with one of the components further down the transmit chain.
 - b. If the DMM measures more than 10 ohms over the value measure in [Step 3](#) then the problem appears to be that the RF Transmit Switch is not switching. Check that the correct control cable is attached to J150 on the RF Transmit Switch and check its route back to J25 on the 16 Channel Switch. Only replace the RF Transmit Switch if everything is connected properly.
 - c. Reconnect the cable to J153
16. Disconnect the Connector A T/R cable at the incoming side of the Cable Interface panel (J2) in the Rear Pedestal, and use a DMM to measure the resistance between the center pin and shell of the cable connector.
 - a. If the DMM measures within 10 ohms of the value recorded in [Step 3](#) check the center pin on the connector. Make sure it is long enough to contact the center conductor in the Cable Interface J2 feed-through connector. The problem may be further down the transmit chain.
 - b. If the DMM measures more than 10 ohms over the value measured in [Step 3](#) then the problem is within the cable. Make certain that this cable is really from J153 on RF Transmit Switch. If cable identity OK then inspect the connectors on each end of the cable and re-terminate the cable ends if necessary.
 - c. Reconnect cable to J2.
17. Disconnect the Head Cable Track cable from the opposite side of Cable Interface J2 in the Rear Pedestal and use a DMM to measure the resistance between the center conductor and shell of the J2 feed-through connector.
 - a. If the DMM measures within 10 ohms of the value recorded in [Step 3](#) check the center pin on the removed cable connector. Make sure it is long enough to contact the center conductor at J2. The problem may be further down the chain.
 - b. If the DMM measures more than 10 ohms over the value measured in [Step 3](#) the problem is with the J2 feed-through connector. Replace the feed-through connector.
 - c. Reconnect cable to J2.

18. Remove the LPCA cover. **Upgrade systems** continue with [Step 19](#), while **Forward production systems** skip to [Step 27](#).
19. Locate Connector A and the transmit cable from pin E1 on Connector A to the bulkhead interface on the LPCA.
20. Disconnect the Connector A T/R cable from the bulkhead interface and use a DMM to measure the resistance between the center pin and shield of the cable connector.
 - a. If the DMM measures within 10 ohms of the value recorded in [Step 3](#) check the center pin on the cable connector. Make sure it is long enough to contact the center conductor in the bulkhead interface. The problem may be with one of the components further down the transmit chain.
 - b. If the DMM measures more than 10 ohms over the value measure in [Step 3](#) then the problem appears to be with the cable. Make certain that this cable is really from J2 in the Rear Pedestal and was not swapped with the Head Transmit cable. If cable identity OK then repair the cable or replace the Cable Track.
 - c. Reconnect the cable to bulkhead interface.
21. Disconnect the Connector A transmit cable from the opposite side of the bulkhead interface and use a DMM to measure the resistance between the center conductor and shell of the bulkhead interface feed-through connector.
 - a. If the DMM measures within 10 ohms of the value recorded in [Step 3](#) check that the cable really does attach to Connector A. If cable identity is OK then repair or replace the Connector A assembly. Refer to the [Replacement of Hypertronics Connector on LPCA](#) procedure.
 - b. If the DMM measures more than 10 ohms over the value measured in [Step 3](#) the problem is with the bulkhead interface feed-through connector. Replace the feed-through connector.
 - c. Return all cables to their proper positions and proceed to the [Finalization, Step 1](#).
22. **(For Upgrade system with failure along Head Bendix connector path:)** Disconnect the Head Transmit cable at J151 on RF Transmit Switch and use a DMM to measure the resistance between the center conductor and shield of J151 on the RF Transmit Switch.
 - a. If the DMM measures within 10 ohms of the value recorded in [Step 3](#) check the center pin on the cable connector. Make sure it is long enough to contact the center conductor in J151 on RF Transmit Switch. The problem may be with one of the components further down the transmit chain.
 - b. If the DMM measures more than 10 ohms over the value measure in [Step 3](#) then the problem appears to be that the RF Transmit Switch is not switching. Replace the RF Transmit Switch.
 - c. Reconnect the cable to J151.
23. Disconnect the Head Transmit cable at the incoming side of the Cable Interface panel J1 in the Rear Pedestal, and use a DMM to measure the resistance between the center pin and shell of the cable connector.
 - a. If the DMM measures within 10 ohms of the value recorded in [Step 3](#) check the center pin on the connector. Make sure it is long enough to contact the center conductor in the Cable Interface J1 feed-through connector. The problem may be further down the transmit chain.
 - b. If the DMM measures more than 10 ohms over the value measured in [Step 3](#) then the problem is within the cable. Make certain that this cable has not been confused with the Connector A T/R cable or the optional MNS transmit cable, if present. If cable identity is OK then repair or replace the cable.
 - c. Reconnect the cable to J1.
24. Disconnect the Head Cable Track cable from the opposite side of Cable Interface J1 in the Rear Pedestal and use a DMM to measure the resistance between the center conductor and shell of the J1 feed-through connector.
 - a. If the DMM measures within 10 ohms of the value recorded in [Step 3](#) check the center pin on the removed cable connector. Make sure it is long enough to contact the center conductor at J2. The problem may be further down the chain.
 - b. If the DMM measures more than 10 ohms over the value measured in [Step 3](#) the problem is with the J1 feed-through connector. Replace the feed-through connector.
 - c. Reconnect cable to J1.

25. Remove the LPCA cover.

If this is a **forward production system** then skip to [Step 27](#).

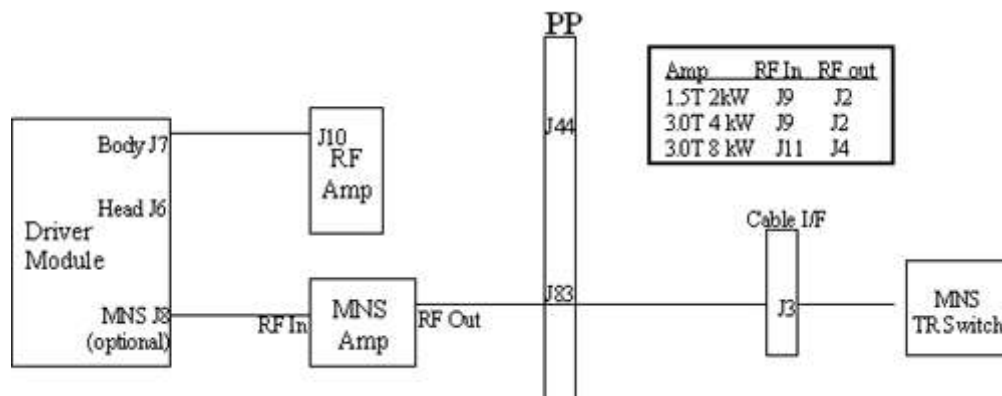
26. **(For Upgrade system:)** Disconnect the Head transmit cable from J6 on the Head TR Network and use a DMM to measure the resistance between the center pin and shell of the cable connector.
 - a. If the DMM measures within 10 ohms of the value recorded in [Step 3](#) check the center pin on the connector. Make sure it is long enough to contact the J6 center conductor. If connector is OK then replace the Head TR Network, refer to the [LPCA Replacements](#) procedure.
 - b. If the DMM measures more than 10 ohms over the value measured in [Step 3](#) then the problem is within the cable. Make certain that this cable has not been confused with the Connector A T/R cable. If cable identity is OK then repair cable or replace the Cable Track, refer to the [Cable Take-up Replacement and Access](#) procedure.
 - c. Return all cables to their proper locations and continue with the [Finalization, Step 1](#).
27. **(For 24. Forward Production:)** Disconnect the Conn A Head transmit cable from J18 on the Multicoil Assembly and use a DMM to measure the resistance between the center pin and shell of the cable connector.
 - a. If the DMM measures within 10 ohms of the value recorded in [Step 3](#) check the center pin on the connector. Make sure it is long enough to contact the J18 center conductor. If the connector looks OK then the problem is further down the transmit chain.
 - b. If the DMM measures more than 10 ohms over the value measured in [Step 3](#) then the problem is within the cable. Replace the Cable Track, refer to the [Cable Take-up Replacement and Access](#) procedure.
 - c. Reconnect cable to J18.
28. Disconnect the Conn A Head transmit cable from J17 on the Multicoil Assembly and use a DMM to measure the resistance between the center conductor and shell of J17 connector.
 - a. If the DMM measures within 10 ohms of the value recorded in [Step 3](#) check the center pin on the cable connector. Make sure it is long enough to contact the J17 center conductor. If the connector looks OK then replace the connector "A" assembly in the front of the LPCA, refer to the [Replacement of Hypertronics Connector on LPCA](#) procedure.
 - b. If the DMM measures more than 10 ohms over the value measured in [Step 3](#) then replace the Multicoil Assembly, refer to the [LPCA Replacements](#) procedure.
29. Reconnect all cables and proceed to the [Finalization, Step 1](#).

4.3 TR Open Circuit Fault – MNS

(For 1.5T upgrade systems with optional MNS: If the MNS TR switch is not plugged in, an MNS TR Open fault will be detected. Please verify an MNS TR switch is plugged in before troubleshooting further.) If the system is reporting a T/R Open Circuit Fault along the MNS line, troubleshooting must occur to isolate it to the failed component. [Illustration 4](#) shows a path for the MNS TR signals.

1. Swap the MNS (J8) and Body (J7) T/R output cables at the Driver Module, and rerun the Diagnostic to determine if the failure changes.
 - a. If the MNS line still reports a T/R Open, and the body does not report any failure, the problem is within the Driver Module. Replace the driver module, refer to the [Driver Module Replacement](#) procedure.
 - b. If the Body path now reports a T/R Open, the problem is somewhere further along the MNS T/R path. Do **NOT** replace the Driver Module – continue to troubleshoot.
 - c. Return the Driver Module MNS and Body T/R output cables to their proper connection points.

Illustration 4: System MNS T/R Signal



2. Swap the MNS Amp (see table in [Illustration 4](#) for J#) and Body (J10) T/R Input cables at the RF Amplifiers, and rerun the Diagnostic to determine if the failure changes.
 - a. If the MNS line still reports a T/R Open, and the body does not report any failure, the problem is within the cable or connection between the Driver Module and MNS RF Amplifier. Verify the connection integrity, and if all connections are good, but still getting an error, replace the cable.
 - b. If the Body path now reports a T/R Open, the problem is somewhere further along the MNS T/R path. Continue to troubleshoot.
 - c. Return the MNS and Body T/R RF Amplifier input cables to their proper connection points.



CAUTION

Strong Magnetic Field!

Ferrous materials can become dangerous projectiles in the presence of the magnetic field Produced by the Signa Magnet.

The Digital Multi Meter (DMM) has ferrous components. Use extreme caution if this tool must be brought into the magnet room, keeping it as far from the magnet as possible.

3. Obtain a 50ohm terminator and use a DMM to measure and record its resistance from center pin to shell.



DANGER

FATAL ELECTRIC SHOCK HAZARD!!

FATAL VOLTAGES FROM THE RF AND MNS AMPLIFIERS WHEN ENTERGIZED.

TO PREVENT FATAL ELECTRIC SHOCK, WHEN DIRECTED, TURN OF THE RFI/AMP AND MNS BREAKERS ON THE FRONT OF THE PDU BEFORE YOU PERFORM THE PROCEDURES. LOCK AND TAG OUT THE PLEXI-GLASS COVER ON THE FRONT OF THE PDU.

4. On the front of the PDU, Lock out and Tag out the MNS Breaker, to remove power from the RF Amplifier.
5. Disconnect the MNS T/R cable at the Driver Module (J8), and connect a 50-ohm terminator to the cable.
6. Disconnect the MNS T/R Switch transmit cable from the SHV connector on the front of the LPCA. See [Illustration 5](#).

Illustration 5: MNS T/R Switch Connection



7. Use a DMM to measure the resistance between the center pin and shield of the SHV connector on the front of the LPCA.
 - a. If the DMM measures within 10 ohms of the value recorded in [Step 3](#) then replace the MNS T/R Switch.
 - b. If the DMM measures more than 10 ohms over the value measured in [Step 3](#) then the problem is further up the chain.
 - c. Reattach the MNS T/R Switch transmit cable.
8. Disconnect the MNS T/R cable at the Cable Interface Panel on the Rear Ped(J3), and use a DMM to measure resistance between the center conductor and shell of the feed-through connector.
 - a. If the DMM measures within 10 ohms of the value recorded in [Step 3](#), the problem is with the cables in the LPCA. Remove the LPCA cover and inspect these cables. Replace and repair as necessary.
 - b. If the DMM measures more than 10 ohms over what was measured in [Step 3](#), the problem is further up the cable path.
 - c. Reconnect the cable to the Cable Interface at J3.
9. Disconnect the MNS T/R cable at the Cable Interface Panel on the opposite side of the Rear Ped(J3), and use a DMM to measure resistance between the center pin and shell of the cable.
 - a. If the DMM measures within 10 ohms of the value recorded in [Step 3](#), then make sure the center pin is long enough to contact the center conductor in the J3 feed-through connector. If center pin OK then replace feed-through connector.
 - b. If the DMM measures more than 10 ohms over what was measured in [Step 3](#), the problem is further up the cable path.
 - c. Reconnect the cable to the Cable Interface at J3.
10. Disconnect the MNS T/R output cable at the Magnet Room side of the Penetration Panel (J83), and use a DMM to measure the resistance between the center conductor and outer shell of the feed-through connector.
 - a. If the DMM measures within 10 ohms of the value recorded in [Step 3](#), the problem is with the cable between the Penetration Panel (J83) and Rear Ped (J3). Check the center pin on the removed cable and make sure it is long enough to contact the center conductor in the feed-through connector. If OK then inspect the connectors on either end of the cable. Repair or replace cable.
 - b. If the DMM measures something more than 10 ohms over what was measured in [Step 3](#), the problem is further down the chain.
 - c. Reconnect the cable to J83 on the Penetration Panel.

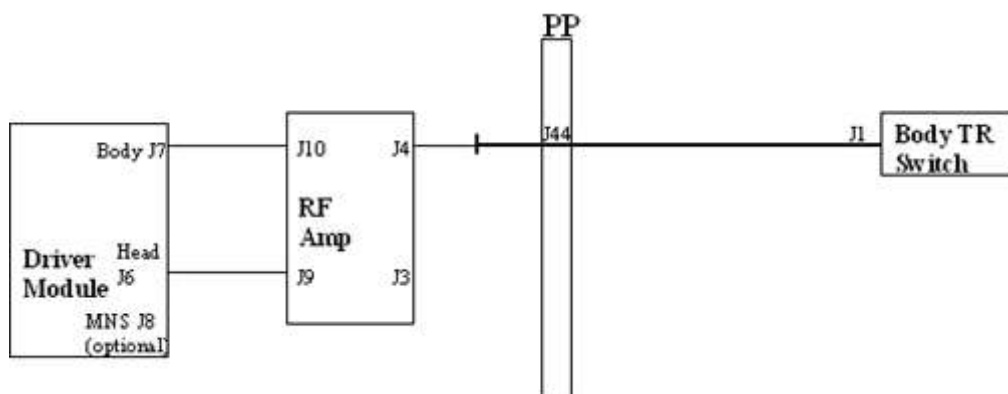
11. Disconnect the MNS T/R output cable at the Equipment Room side of the Penetration Panel (J83), and use a DMM to measure the resistance between the center pin and outer shell of the connector.
 - a. If the DMM measures within 10 ohms of the value recorded in [Step 3](#), check the center pin in the cable and make sure it is long enough to contact the center conductor in the J83 feed-through connector. If OK the replace the feed-through connector.
 - b. If the DMM measures something more than 10 ohms over what was measured in [Step 3](#), the problem is further down the chain.
 - c. Reconnect the cable to J83 on the Equipment room side of the Pen Panel.
12. Disconnect the MNS T/R output cable at the RF Amp MNS RF Out (see table in [Illustration 4](#) for correct J#), and use a DMM to measure the resistance between the center conductor and outer shell of the J2 connector on the Amp.
 - a. If the DMM measures within 10 ohms of the value recorded in [Step 3](#), the problem is with the cable between the RF Amp MNS RF Out and the Penetration Panel J83. Check the center pin on the removed cable and make sure it is long enough to contact the center conductor on RF Amp MNS RF Out port. If OK then check connectors on each end of cable. Repair or replace cable.
 - b. If the DMM measures something more than 10 ohms over what was measured in [Step 3](#), the problem is with the MNS RF Amplifier. Recheck the connector and center pin at the RF In port (see table in [Illustration 4](#) for correct J#) on the MNS Amp, and if still getting the fault, repair or replace the MNS Amplifier. Refer to the [3.0T CPC MNS Amplifier Replacement](#) procedure.
 - c. Remove the 50-ohm terminator, and reconnect the cable to the Driver Module and at the Amplifier.

4.4 TR Short Circuit Fault – Body

If the system is reporting a T/R Short Circuit Fault along the Body path, troubleshooting must occur to isolate it to the failed component. [Illustration 6](#) shows a path for the Body TR signals.

1. Disconnect at Driver Module (J7) and repeat the Diagnostic.
 - a. If still get a short, then issue is in the Driver Module. Replace the Driver Module, refer to the [Driver Module Replacement](#) procedure.
 - b. If only get an open circuit fault (due to the disconnected cable), then the short is further along the path past the induced open, and is **NOT** the Driver Module. Continue to troubleshoot.
 - c. Reconnect the cable at J7 on the Driver Module.

Illustration 6: System Body T/R Signal Path



2. Disconnect at the RF Amp (J10) input cable and repeat the Diagnostic.
 - a. If still get a short, then issue is in the cable between the Driver Module and the RF Amplifier. Inspect the cable and connectors, and repair or replace the cable.
 - b. If only get an open circuit fault (due to the disconnected cable), then short is further along the path past the induced open. Continue to troubleshoot.

- c. Reconnect the cable at J10 on the Amplifier.
3. Disconnect at the RF Amp (J4) output and repeat the Diagnostic.
 - a. If still get a short, reinspect cable at J10 for an internal short condition only seen when connected. Otherwise then issue is in the RF Amplifier. Continue to troubleshoot the Amplifier, and if necessary, replace the Amplifier according to replacement instructions.
 - b. If only get an open circuit fault (due to the disconnected cable), then short is further along the path past the induced open. Continue to troubleshoot.
 - c. Reconnect the cable at J4 of the Amplifier.
4. Disconnect at the Body Hybrid (J1) in the Rear Pedestal and repeat the Diagnostic.
 - a. If still get a short, then issue is in the cable chain between Amp and Hybrid. Inspect the cable connectors and the Pen Panel connectors. Repair or replace any damaged cables.
 - b. If the short goes away, and only get an Open circuit fault, then problem is most likely the Body Hybrid/ TR Switch. Replace the Body Hybrid, refer to the following procedure.
 - c. Reconnect the cable at J1 on the Body Hybrid.

4.5 TR Short Circuit Fault – Head

If the system is reporting a T/R Short Circuit Fault along the Head path, troubleshooting must occur to isolate it to the failed component. Illustration 7 and 8 show a path for the Head TR signals. There are differences between an upgraded system, which still has the Legacy connector, and a forward production system with the new LPCA. An upgraded system utilizes a RF TX switch on the inside of the Pen-Panel to split the head signal into two. One supplies the Head Port, while the other supplies Connector A. In a forward production system, the splitter is bypassed, so there is only one head signal in the Magnet room.

1. **(For Upgrade systems:)** if getting a short on the Head path, Un-plug the coil in the Legacy port and repeat the Diagnostic, to verify it is not a short in the coil.

NOTE: For a Conn A Head Path Short Circuit Fault, unplugging the coil will not work, as the test will only check for a short if a T/R coil is plugged in. Continue with [Step 2](#) in this situation, continuing to keep the coil plugged in.

- a. If still get the error, then short is in the system path, and is **NOT** the coil.
- b. If no longer getting the short circuit fault, the issue is within the coil.

Illustration 7: Upgrade System Head T/R Signal Path

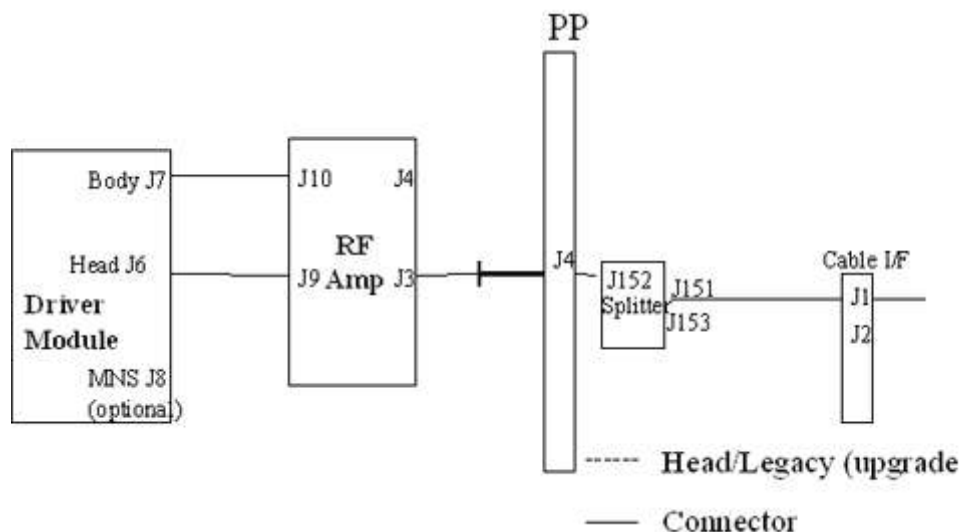
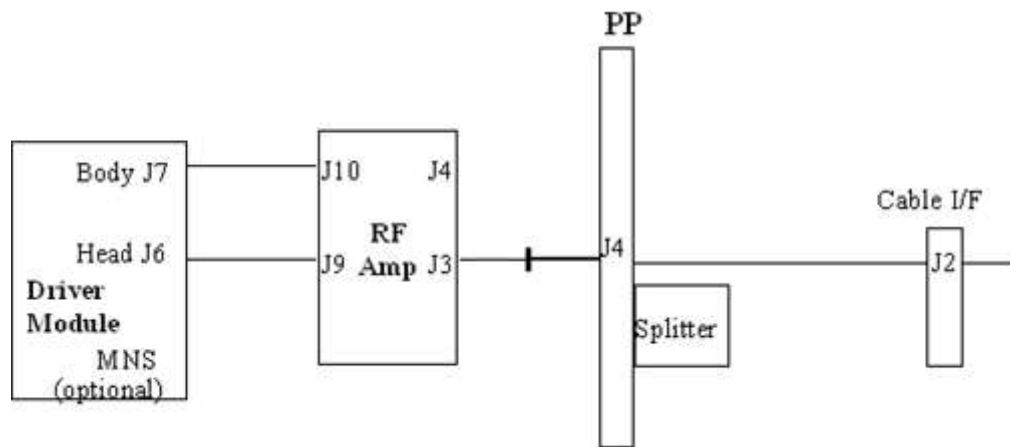


Illustration 8: Forward Production Head T/R Signal Path



Start disconnecting cables along the path to determine where the short is located.

2. Disconnect at Driver Module (J6) and repeat the Diagnostic.

- If still get a short, then issue is in the Driver Module. Replace the Driver Module, refer to the [Driver Module Replacement](#) procedure.
- If only get an open circuit fault (due to the disconnected cable), then short is further along the path, past the open, and is **NOT** the Driver Module. Continue to troubleshoot.
- Reconnect the cable at J6.

3. Disconnect at the RF Amp (J9) input and repeat the Diagnostic.

- If still get a short, then issue is in the cable between the Driver Module and the RF Amplifier. Replace the cable.
- If only get an open circuit fault (due to the disconnected cable), then short is further along the path past the open. Continue to troubleshoot.
- Reconnect the cable at J9.

4. Disconnect at the RF Amp (J3) output and repeat the Diagnostic.

- If still get a short, reinspect cable at J9 for internal short condition only seen when connected. Otherwise replace the Amplifier according to replacement instructions.
- If only get an open circuit fault (due to the disconnected cable), then short is further along the path past the induced open. Continue to troubleshoot.
- Reconnect the cable at J3 of the RF Amp.

5. **(For Connector A Head Path short:)** Disconnect at the Cable Interface Panel (J2) on the Rear-Pedestal and repeat the Diagnostic.

- If still get a short, then issue is in the cable chain between Amp and Cable I/F Panel.
- If the short goes away, and only get an Open circuit fault, then problem is inside the LPCA. Continue troubleshooting these components.
- Reconnect the cable at J2.

6. **(For Head Path short, Upgrade systems only:)** Disconnect at the Cable Interface Panel (J1) at the Rear-Pedestal and repeat the Diagnostic.

- If still get a short, then issue is in the cable chain between Amp and Cable I/F Panel.
- If the short goes away, and only get an Open circuit fault, then problem is within the LPCA cables or port. Continue troubleshooting these components, and replace any failed parts.
- Reconnect the cable at J1.

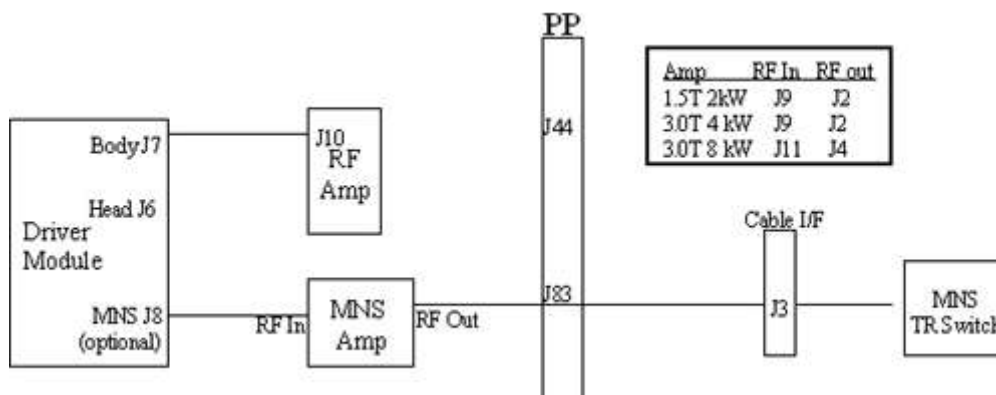
7. On the Equipment side of the Pen-panel disconnect J4 and repeat the Diagnostic.
 - a. If still get a short, then issue is in the cable between Amp and Pen-panel. Inspect the cable and connectors, and repair or replace the cable.
 - b. If the short goes away, and only get an Open circuit fault, then problem is either the cable in the magnet room, between the pen-panel and the Cable I/F Panel, or the RF TX Switch & cables. If this is a **forward production system**, replace the cable. For an **upgrade system**, continue with [Step 8](#).
 - c. Reconnect the cable at J4.
8. **(For Head Path short, Upgrade systems only:)** disconnect at inside of the Pen-panel, at the output of the RF TX Switch (J151) and repeat the Diagnostic.
 - a. If still get a short, then issue is in the RF TX Switch or Pen-Panel cabling. Verify cable connections and replace the RF TX Switch if necessary.
 - b. If the short goes away, and only get an Open circuit fault, then problem is the cable in the magnet room, between the pen-panel and the Cable I/F Panel. Replace the cable.
 - c. Reconnect the cable at J151.

4.6 TR Short Circuit Fault – MNS

If the system is reporting a T/R Short Circuit Fault along the MNS line, troubleshooting must occur to isolate it to the failed component. [Illustration 9](#) shows a path for the TR signals.

1. Un-plug the MNS Coil and repeat the Diagnostic, to verify it is not a short in the coil.
 - a. If still get the error, then short is in the system path, and is **NOT** the coil.
 - b. If no longer getting the short circuit fault, the issue is within the MNS coil.

Illustration 9: System MNS T/R Signal



Start disconnecting cables along the path to determine where the short is located.

2. Disconnect at Driver Module (J8) and repeat the Diagnostic.
 - a. If still get a short, then issue is in the Driver Module. Replace the Driver Module, refer to the [Driver Module Replacement](#) procedure.
 - b. If only get an open circuit fault (due to the disconnected cable), then short is further along the path past the open, and is **NOT** the Driver Module. Continue to troubleshoot.
 - c. Reconnect the cable at J8 of the Driver Module.
3. Disconnect at the MNS RF Amp (See table in [Illustration 9](#) for J#) input and repeat the Diagnostic.
 - a. If still get a short, then issue is in the cable between the Driver Module and the MNS RF Amplifier. Replace the cable.

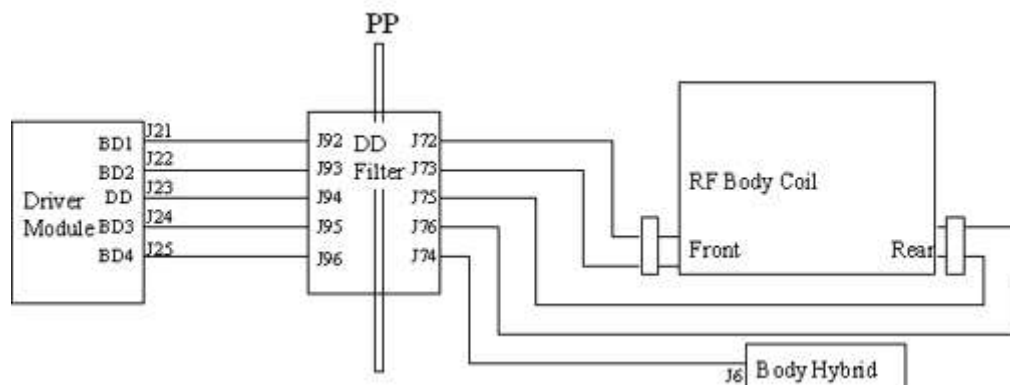
- b. If only get an open circuit fault (due to the disconnected cable), then short is further along the path past the induced open. Continue to troubleshoot.
 - c. Reconnect the cable at the MNS Amp input.
 4. Disconnect at the MNS RF Amp (See table in [Illustration 9](#) for J#) output and repeat the Diagnostic.
 - a. If still get a short, reinspect MNS Amp input cable for internal short condition only seen when connected. If no issues are found with the cable, replace the Amplifier according to replacement instructions.
 - b. If only get an open circuit fault (due to the disconnected cable), then short is further along the path past the induced open. Continue to troubleshoot.
 - c. Reconnect the cable at RF output of the MNS Amplifier.
 5. Disconnect at the Cable I/F Panel (J3) on the Rear Ped. and repeat the Diagnostic.
 - a. If still get a short, then issue is in the cable chain between the MNS Amp and the I/F Panel. Continue with [Step 6](#).
 - b. If the short goes away, and only get an Open circuit fault, then problem is either the MNS TR Switch, or cabling in the LPCA. Continue troubleshooting these components, and replace the failed parts as needed.
 - c. Reconnect the cable at J3.
 6. Disconnect from the Equipment Room side of the Pen-panel (J83) and repeat the Diagnostic.
 - a. If still get a short, then issue is in the cable between MNS Amp and Pen-panel. Replace the cable.
 - b. If the short goes away, and only get an Open circuit fault, then problem is most likely the cable in the magnet room, between the pen-panel and the Cable Interface of the Rear Pedestal. Replace the cable.
 - c. Reconnect the cable at J83.

4.7 Dynamic Disable and Direct Drive High Voltage Fault

If the system is reporting a DD High Voltage Fault, troubleshooting must occur to isolate it to the failed component. The error should give information specific to which path has the issue. If all lines do not have voltage within bounds, then the system should give an error, and suggest checking that the HV Switch on the Driver Module interface panel is not in the **Disable** position. [Illustration 10](#) shows the DD signal paths.

1. Disconnect all DD cables from the Driver Module J21-J25, and rerun the diagnostic.
 - a. If still get a High Voltage fault, then the issue is a component within the Driver Module. Replace the Driver Module, refer to the [Driver Module Replacement](#) procedure.
 - b. If the High Voltage fault goes away, continue with [Step 2](#) for troubleshooting.

Illustration 10: DD System Paths



2. Reattach cables one at a time, and run the diagnostic at each point to verify which DD path is pulling the supply down. Based on the DD path that is reporting the fault, continue with [Step 3](#).

3. Swap the problem path on the Magnet room side of Pen-Panel at DD Filter with the cables as follows, and rerun the diagnostic. If BD1-BD4 (J72, J73, J75, J76) reports error, swap with DD (J74). If DD (J74) reports an error, swap with BD4 (J76).
 - a. If continue to see the HV Fault on the original BDx (x=1-4) or DD line, then the issue is the DD Filter Board or cable between the Driver Module and Pen Panel.
 - i. Return the swapped cables to their appropriate connection points.
 - ii. Swap the corresponding cables on the Equipment Side the Pen-Panel and run the test. If the HV Fault does not move, the issue is the cable connecting the Pen-panel to the Driver module. Inspect cable and connectors, and repair or replace the cable.
 - iii. If the HV Fault moves to the other path, the issue is the Filter box on the Pen-Panel. Replace the Filter box.
 - b. If HV Fault moves to the swapped line, the fault is further down the path. Continue following the troubleshooting steps.
 - c. Return the swapped cables to their appropriate connection points on the Filter Board.
4. Swap the cable end of the Problem Path (Run 778,779,781,782) with the DD line (Run 780) at the rear pedestal. If the DD line (Run780) is the fault path, swap it with BD4 (Run782). Rerun the diagnostic.
 - a. If system continues to report a High Voltage fault on the original DD line, then problem is the cable between the Pen-panel and the cable I/F at the back of the magnet. Inspect the cable and connectors, and repair or replace the cable.
 - b. If the High Voltage fault moves to the swapped path, problem is something with the Body coil or Body DD Boards. Continue with the troubleshooting steps.
 - c. Return the swapped cables to their appropriate connections at the back of the magnet.
5. Follow the [Dynamic Disable Switch Board Replacement](#) procedure and check the DD board diodes with a DMM for the DD Board in question. Voltage should be 0.35-0.6V in one direction, open in the other.
 - a. If the Voltage is not within range, replace the DD Board.
 - b. If the Voltage is within the specified range, contact Support for more assistance in troubleshooting the Body Coil.

4.8 DD Open Circuit Fault

If the system is reporting a DD Open Circuit Fault, troubleshooting must occur to isolate it to the failed component. The system will specify which DD path is experiencing the open. Illustration 10 shows a path for the DD circuit.

1. Swap the problem path at the Driver Module with cable suggested below, and rerun the diagnostic. If BD1-BD4 (J72, J73, J75, J76) reports error, swap with DD (J74). If DD (J74) reports an error, swap with BD4 (J76).
 - a. If still continue to see the open on the original BDx (x=1-4) or DD line, then the issue is with the driver module. Replace the driver Module, refer to the [Driver Module Replacement](#) procedure.
 - b. If open moves to the swapped line, the open is further down the path. Continue following the troubleshooting steps.
 - c. Return the swapped cables to their appropriate connection points on the Driver Module.
2. Swap the problem cable on the Magnet Side of Pen-Panel at Filter board with the cables suggested below, and rerun the diagnostic. If BD1-BD4 (J72, J73, J75, J76) reports error, swap with DD (J74). If DD (J74) reports an error, swap with BD4 (J76).
 - a. If still continue to see the open on the original BDx (x=1-4) or DD line, then the issue is the DD Filter Board or cable.
 - i. Return the swapped cables to their appropriate connection points.

- ii. Swap the corresponding cables on the Equipment Room side of the Pen-Panel and run the test. If the open does not move, the issue is the cable connecting the Pen-panel to the Driver module. Inspect the cable and connectors, and repair or replace the cable.
 - iii. If the Open moves to the other path, the issue is the Filter box on the Pen-Panel. Replace the Filter box, refer to the [Gradient Filter Replacement Procedure](#).
- b. If open moves to the swapped line, the open is further down the path. Continue following the troubleshooting steps.
- c. Return the swapped cables to their appropriate connection points on the Filter Box.
- 3. Select the problem cable from the bullets below, and swap the end of the cable at the Rear Pedestal with the cable end suggested and rerun the diagnostic.
 - BD1 (Run 778, J3) with DD (Run 780)
 - BD2 (Run 779, J4) with DD (Run 780)
 - BD3 (Run 781, J5) with DD (Run 780)
 - BD4 (Run 782, J6) with DD (Run 780)
 - DD (Run 780) with BD4 (Run 782, J6)
- 4. If continue to see the open on the original BD path, then issue is the cable between the Pen Panel, and the cable interfaces. Inspect the cable and connectors, and repair or replace the suspect cable.
 - a. If For BD1-BD4 issues, follow the [Dynamic Disable Switch Board Replacement](#) procedure and check the DD board diodes with a DMM for the DD Board in question. Voltage should be 0.35-0.6V in one direction, open in the other.
 - b. If the Voltage is not within range, replace the DD Board.
 - c. If the Voltage is within the specified range, contact Support for more assistance in troubleshooting the Body Coil.
- 5. Return the cables to their original locations.

4.9 DD Short Circuit Fault

If the system is reporting a DD Short Circuit Fault, troubleshooting must occur to isolate it to the failed component. The system will specify which DD path is experiencing the short. [Illustration 10](#) shows paths for the DD circuits.

1. Start disconnecting cables along the path to determine where the short is located.
2. Disconnect the faulty line at the Driver Module (J21-J25)) and repeat the Diagnostic.
 - a. If still get a short, then issue is in the Driver Module. Replace the Driver Module, refer to the [Driver Module Replacement](#) procedure.
 - b. If only get an open circuit fault (due to the disconnected cable), then short is further along the path past the open, and is **NOT** the Driver Module. Continue to troubleshoot.
 - c. Reconnect the cable at the Driver Module.
3. Disconnect at the DD Filter on the Equipment Room side of the Pen-panel (J92-J96), and repeat the Diagnostic.
 - a. If still get a short, then issue is in the cable between the Driver Module and the Pen-panel. Inspect the cable and connectors, and repair or replace the cable.
 - b. If only get an open circuit fault (due to the disconnected cable), then short is further along the path past the induced open. Continue to troubleshoot.
 - c. Reconnect the cable at the Pen-panel.
4. Disconnect at the DD Filter output on the Magnet Room side of the Pen-panel (J72-J76) and repeat the Diagnostic.

- a. If still get a short, then issue is in the DD Filter. Replace the Filter.
 - b. If only get an open circuit fault (due to the disconnected cable), then short is further along the path past the induced open. Continue to troubleshoot.
 - c. Reconnect the cable at the Pen-panel.
5. **(For Direct Drive Fault:)** Disconnect at the Body Hybrid (J6) and repeat the Diagnostic.
- a. If still get a short, then issue is in the cable between the Pen-panel and Hybrid. Inspect the cable and connectors, and repair or replace the cable.
 - b. If the short goes away, and only get an Open circuit fault, then problem is most likely the Body Hybrid/ TR Switch. Replace the Body Hybrid according to the Replacement instructions.
 - c. Reconnect the cable at the Body Hybrid.
6. **(For BD (Dynamic Disable) Faults:)** Disconnect the cable at the cable interface before it goes into the bore.
- a. If still get a short, then issue is in the cable between Pen-panel and the interface. Inspect the cable and connectors, and repair or replace the cable.
 - b. If the short goes away, and only get an Open circuit fault, then problem is most likely with the Body DD Boards on Body coil.
 - i. Follow the [Dynamic Disable Switch Board Replacement](#) procedure and check the DD board diodes with a DMM (in diode drop mode) for the DD Board in question. Voltage drop should be 0.35-0.6V in one direction, open in the other.
 - ii. If the voltage drop is not within range, replace the DD Board.
 - iii. If the voltage drop is within the specified range, contact Support for more assistance in troubleshooting the Body Coil.
 - c. Reconnect the cable at the cable Interface.

4.10 MC Open Circuit Fault

If the system is reporting a MC Open Circuit Fault, troubleshooting must occur to isolate it to the failed component. [Illustration 11](#) shows a path for the MC signals from the Driver Module through the 16-channel switch. [Table 2](#) is also helpful to understand how the signals flow. Use [Table 2](#) to find your system type, and the Coil Port you are experiencing the problem with. Select the link in the Coil port field to be taken to troubleshooting steps specific to that issue.

NOTE: A coil must be connected in the “A”, “B”, or “C” connectors before the system will check for an MC Open Circuit Fault on these connectors. This is not true for the legacy Bendix connector as it is checked even with no coil attached. Please verify an MC coil is available before troubleshooting the “A”, “B”, or “C” connectors. Use the one that was in use when the error originally occurred. Review the [HDx Surface Coils](#) document for information on applicable coils.

1. **Illustration 11: System MC Circuit Path**

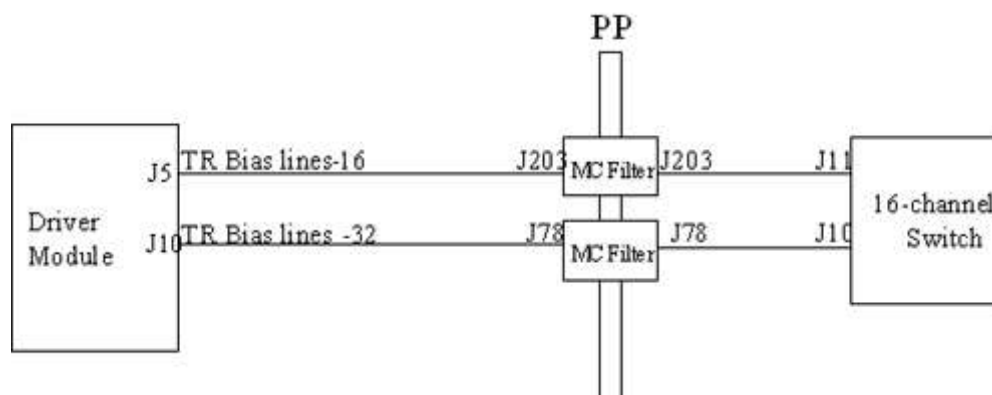


Table 2: System MC Circuit Path

System Type	Channel #	Coil Ports	Driver Module	Pen Panel/MC Filter	16-ch Switch	I/R Filter Box	Coil Connector
Upgrade	8ch & 16ch	Bendix Ch 1-8, MCBias 17-24, refer to Upgrade System, Fault on MC Bendix Connector	J10 DM to J78 Pen Panel	J78 Pen Panel to J10 on 16ch Switch	J19 16ch Switch to J9 Multicoil Assy	N/A	Multicoil Assy to Connector
		Connector A Ch 1-8, MCBias 1-8, refer to Upgrade System, Fault on Connector A & B only	J5 DM to J203 Pen Panel	J203 Pen Panel to J11 on 16ch Switch	J18 16ch Switch to J12 LPCA Bulkhead	N/A	J12 Bulkhead to Connector Port
		Connector B Ch 1-8, MCBias 9-16, refer to Upgrade System, Fault on Connector A & B only	J5 DM to J203 Pen Panel	J203 Pen Panel to J11 on 16ch Switch	J17 16ch switch J22 LPCA Bulkhead	N/A	J22 Bulkhead to Connector Port
Forward Prod	8ch & 16ch	Connector A Ch 1-8 MCBias 1-8, refer to Forward Production, Fault on Connector A	J5 DM to J203 Pen Panel	J203 Pen Panel to J11 of 16ch Switch	J18 16-ch switch to J21 MC I/F Box	MC Interface box to Connector	MC Interface box to Connector
		Connector B Ch 1-8 MCBias 9-16, refer to Forward Production, Fault on Connector B, Coil Ch 1-8 (Bias Ch 9-16) ONLY	J5 DM to J203 Pen Panel	J203 Pen Panel to J11 of 16ch Switch	J17 16ch switch to J6 Image Reject Filter box	Image Reject Filter box J1 to LPCA Bulkhead	LPCA Bulkhead to Connector Port
	16ch	Connector B Ch 9-16 MCBias 17-24, refer to Forward Production, Fault on Connector B, Coil Ch	J10 DM to J78 Pen Panel	J78 Pen Panel to J10 of 16ch Switch	J19 16ch switch to J8 Image Reject Filter box	Image Reject Filter box J3 to LPCA Bulkhead	LPCA Bulkhead to Connector Port

System Type	Channel #	Coil Ports	Driver Module	Pen Panel/MC Filter	16-ch Switch	I/R Filter Box	Coil Connector
		9-16 (Bias Ch 17-24) ONLY					

4.10.1 Upgrade System, Fault on MC Bendix Connector

- Disconnect coil from the Bendix connector and run the MC TR DD diagnostic.
 - If the fault is gone then the coil has failed. Check the coil cable connector. Repair or replace the coil.
 - If the fault remains then the problem is further down the chain and the coil is OK.
 - Leave the coil disconnected from the Bendix connector for the remainder of the troubleshooting process.
- Run the diagnostic and note the failing channel.
 - Open a c-shell window, size it the width of the screen, and in the window type `more /usr/g/service/log/System_Path*`. Scroll down to the bottom of the file and find the data with the diagnostic execution time corresponding to the last run of the diagnostic. See [Illustration 12](#).

Illustration 12: System_Path_Diag_Log.txt Output

Diagnostic Execution Time: SAT FEB 11 16:14:18 2005

LPTA Type: Upgrade System Field Strength: 1.5 T

System NOT Configured for HMI

Connector Multicoil Lower: No Coil Detected
 Connector Multicoil A: No Coil Detected
 Connector Multicoil B: No Coil Detected

MC Data

Channel	Connector	Short Ckt Current	Short Ckt Limit	Open Ckt Current	Open Ckt Limit	Short Status	Open Status
1	A1	0	-50	0	0	No Fault	Not Checked
2	A2	0	-50	0	0	No Fault	Not Checked
3	A3	0	-50	0	0	No Fault	Not Checked
4	A4	0	-50	0	0	No Fault	Not Checked
5	A5	0	-50	0	0	No Fault	Not Checked
6	A6	0	-50	0	0	No Fault	Not Checked
7	A7	0	-50	0	0	No Fault	Not Checked
8	A8	0	-50	0	0	No Fault	Not Checked
9	B1	0	-50	0	0	No Fault	Not Checked
10	B2	0	-50	0	0	No Fault	Not Checked
11	B3	0	-50	0	0	No Fault	Not Checked
12	B4	0	-50	0	0	No Fault	Not Checked
13	B5	0	-50	0	0	No Fault	Not Checked
14	B6	0	-50	0	0	No Fault	Not Checked
15	B7	0	-50	0	0	No Fault	Not Checked
16	B8	0	-50	0	0	No Fault	Not Checked
17	L1	-120	-170	181	230	No Fault	No Fault
18	L2	-131	-170	155	230	No Fault	No Fault
19	L3	-123	-170	163	230	No Fault	No Fault
20	L4	-131	-170	155	230	No Fault	No Fault
21	L5	-118	-170	164	230	No Fault	No Fault
22	L6	-131	-170	161	230	No Fault	No Fault
23	L7	-120	-170	164	230	No Fault	No Fault
24	L8	-122	-170	160	230	No Fault	No Fault
25	E	0	-50	0	0	No Fault	Not Checked
26	F	0	-50	0	0	No Fault	Not Checked
27	G	0	-50	0	0	No Fault	Not Checked
28	H	0	-50	0	0	No Fault	Not Checked
29	I	0	-50	0	0	No Fault	Not Checked
30	J	0	-50	0	0	No Fault	Not Checked
31	K	0	-50	0	0	No Fault	Not Checked
32	L	0	-50	0	0	No Fault	Not Checked

MC bias currents for legacy Bendix connector channels.

- Find the fault in Open or Short status column on the right and then match that with a channel on the left. In the Open Ckt Current or Short Ckt Current column note the failing current value.
- Swap the connectors at J5 and J10 on the rear of the Driver Module and run the diagnostic again.
 - Scroll down the c-shell window to find the most recent data.
 - Swapping the cables means that the MC bias lines have shifted down 16 places to channels 1-8. Channels 17-24 are now connected to an open coil connector. If the failure was on channel 17 before it could now be on channel 1 and if it was on channel 24 before it could now be on channel 18. Ignore all the new failures now reported in the Open Status and Short Status columns, and any failures shown on the Diagnostic GUI.
 - If the failing current value is now present on one of the channels between 1 and 8 then the problem is further down the chain and the Driver Module is OK.
 - If the failing current value is still present on one of the channels between 17 and 24 then the problem is inside the Driver Module. Repair or replace the Driver Module, refer to the [Driver Module Replacement](#) procedure.

- c. Return the connectors at J5 and J10 on the rear of the Driver Module to the normal positions. Close the c-shell window.
6. Swap the TR Bias cable running between J5 of the Driver Module and J78 of the Pen Panel with the cable running between J10 of the Driver Module and J203 of the Pen Panel. Then run the diagnostic again, with the coil connected to the original port that reported the fault.
 - a. If there is no longer a fault, the open could be due to a faulty cable or connection between the Driver Module and Pen Panel. Inspect the cable and connectors of the cable now connected to J5 of the Driver Module and repair or replace as necessary.
 - b. If the fault continues to show up on the same path, this cable is not the problem. Continue troubleshooting.
 - c. Return the cables to their proper connection points at the Driver Module and Pen Panel.
7. Swap the connection J78 with J203 on the Equipment side of the Pen Panel. Then also swap the connection J78 and J203 on the Magnet room side of the Pen Panel. This will allow the signal to go through the other MC Filter at the Pen Panel to determine if that is the trouble area. Run the diagnostic again.
 - a. If there is no longer a fault, the open could be due to a faulty MC Filter box at the Pen Panel (J78 one), or the cable connections at this box. Inspect the connectors on the MC Filter, and repair or replace the MC Filter box.
 - b. If the fault continues to show up on the same path, the MC filter box is not the problem. Continue troubleshooting.
 - c. Return the cables to their proper connection points on both the Equipment room and Magnet room sides of the Pen Panel.
8. Swap the TR Bias cable running between J203 of the Magnet room side of the Pen Panel, with J11 of the 16-channel switch, with the cable running between J78 of the Pen Panel, and J10 of the 16-channel switch. Run the diagnostic again.
 - a. If there is no longer a fault, the open could be due to a faulty cable or connection between the Pen Panel and the 16-channel switch. Inspect the cable and connections of the cable now connected to J203 of the Pen Panel and repair or replace as necessary.
 - b. If the fault continues to show up on the same path, this cable is not the problem. Continue troubleshooting.
 - c. Return the cables to their proper connection points at the Pen Panel and 16-channel switch.
9. Swap the preamp inside the Multicoil Integrated Assembly on the failing channel with one next to it. Run the diagnostic.
10. If the failure moves to the other channel the replace the failing preamp, refer to the [Multicoil Preamp Replacement](#) procedure.
11. If the failure does not move then order the empty Multicoil Integrated Assembly box and the cable-track cable. Temporarily route the cable between the 16 Channel Switch and Multicoil Integrated Assembly box first to test before replacing the Multicoil Assembly box. The cable is identical in construction to a standard 25-pin serial cable and pin connections are one-to-one from one end of the cable to the other.

4.10.2 Upgrade System, Fault on Connector A & B only

NOTE: The troubleshooting steps in this section require the MCR2 tool. **Upgraded systems** should only use the MCR2 Tool.

1. Remove the coil and connect the MCR2 Tool to the failing connector. Run the TR DD MC diagnostic.
 - a. If fault still on the original MC line, then the failure is not in the coil but further down the signal path.
 - b. If the fault is gone then the coil appears faulty. Check coil connector, and replace or repair coil.
2. Open a c-shell window, size it the width of the screen, and in the window type `more /usr/g/service/log/System_Path*`. Scroll down to the bottom of the file and find the data with the diagnostic execution time corresponding to the last run of the diagnostic. See [Illustration 13](#).

Illustration 13: System_Path_Diag_Log.txt Output

Diagnostic Execution Time: SAT FEB 11 14:28:14 2011

LFPA Type: Upgrade System Field Strength: 1.5 T

System NOT Configured for HSE

Coil Bias Coil ID: coilname shown

Actual open-circuit and short-circuit currents shown in milliamperes (mA)

MC Data

Channel	Connector	Short Ckt Current	Short Ckt Limit	Open Ckt Current	Open Ckt Limit	Short Status	Open Status
1	A1	5	-50	0	0	No Fault	Not Checked
2	A2	5	-50	0	0	No Fault	Not Checked
3	A3	5	-50	0	0	No Fault	Not Checked
4	A4	5	-50	0	0	No Fault	Not Checked
5	A5	5	-50	0	0	No Fault	Not Checked
6	A6	5	-50	0	0	No Fault	Not Checked
7	A7	5	-50	0	0	No Fault	Not Checked
8	A8	5	-50	0	0	No Fault	Not Checked
9	B1	5	-50	0	0	No Fault	Not Checked
10	B2	5	-50	0	0	No Fault	Not Checked
11	B3	5	-50	0	0	No Fault	Not Checked
12	B4	5	-50	0	0	No Fault	Not Checked
13	B5	5	-50	0	0	No Fault	Not Checked
14	B6	5	-50	0	0	No Fault	Not Checked
15	B7	5	-50	0	0	No Fault	Not Checked
16	B8	5	-50	0	0	No Fault	Not Checked
17	L1	-120	-170	152	210	No Fault	No Fault
18	L2	-112	-170	159	210	No Fault	No Fault
19	L3	-117	-170	153	210	No Fault	No Fault
20	L4	-132	-170	159	210	No Fault	No Fault
21	L5	-110	-170	164	210	No Fault	No Fault
22	L6	-124	-170	162	210	No Fault	No Fault
23	L7	-120	-170	164	210	No Fault	No Fault
24	L8	-132	-170	159	210	No Fault	No Fault
25	L9	5	-50	0	0	No Fault	Not Checked
26	L10	5	-50	0	0	No Fault	Not Checked
27	L11	5	-50	0	0	No Fault	Not Checked
28	L12	5	-50	0	0	No Fault	Not Checked
29	L13	5	-50	0	0	No Fault	Not Checked
30	L14	5	-50	0	0	No Fault	Not Checked
31	L15	5	-50	0	0	No Fault	Not Checked
32	L16	5	-50	0	0	No Fault	Not Checked
33	L17	5	-50	0	0	No Fault	Not Checked

MC bias currents for A and B connector channels.

3. Find the fault in Open or Short status column on the right and then match that with a channel on the left. In the Open Ckt Current or Short Ckt Current column note the failing current value.
4. Swap the connectors at J5 and J10 on the rear of the Driver Module and run the diagnostic again.
 - a. Scroll down the c-shell window to find the most recent data.
 - b. Swapping the cables means that the MC bias lines for the coil have shifted up 16 places. This means that if the failure was on channel 5 before it could now be on channel 21 and if it was on channel 14 before it could now be on channel 30. Ignore all the new failures now reported in the Open Status and Short Status columns.
 - c. If the failing current value is now present on one of the channels between 17 and 32 then the problem is further down the chain and the Driver Module is OK.
 - d. If the failing current value is not seen on one of the channels between 17 and 32 then the problem is inside the Driver Module. Repair or replace the Driver Module, refer to the [Driver Module Replacement](#) procedure.
 - e. Return the connectors at J5 and J10 on the rear of the Driver Module to the normal positions. Close the c-shell window.
5. Swap the TR Bias cable running between J5 of the Driver Module and J78 of the Pen Panel with the cable running between J10 of the Driver Module and J203 of the Pen Panel. Then run the diagnostic again, with the coil connected to the original port that reported the fault.
 - a. If there is no longer a fault, the open could be due to a faulty cable or connection between the Driver Module and Pen Panel. Inspect the cable and connections of the cable now connected to J10 of the Driver Module and repair or replace as necessary.
 - b. If the fault continues to show up on the same path, this cable is not the problem. Continue troubleshooting.
 - c. Return the cables to their proper connection points at the Driver Module and Pen Panel.
6. Swap the connection J78 with J203 on the Equipment side of the Pen Panel. Then also swap the connection J78 and J203 on the Magnet room side of the Pen Panel. This will allow the signal to go through the other MC Filter at the Pen Panel to determine if that is the trouble area. Run the diagnostic again.
 - a. If there is no longer a fault, the open could be due to a faulty MC Filter box at the Pen Panel (J203 one), or the cable connections at this box. Inspect the connectors on the MC Filter, and repair or replace the MC Filter box.

- b. If the fault continues to show up on the same path, the MC filter box is not the problem. Continue troubleshooting.
 - c. Return the cables to their proper connection points on both the Equipment room and Magnet room sides of the Pen Panel.
- 7. Swap the TR Bias cable running between J203 of the Magnet room side of the Pen Panel, with J11 of the 16-channel switch, with the cable running between J78 of the Pen Panel, and J10 of the 16-channel switch. Run the diagnostic again.
 - a. If there is no longer a fault, the open could be due to a faulty cable or connection between the Pen Panel and the 16-channel switch. Inspect the cable and connections of the cable now connected to J78 of the Pen Panel and repair or replace as necessary.
 - b. If the fault continues to show up on the same path, this cable is not the problem. Continue troubleshooting.
 - c. Return the cables to their proper connection points at the Pen Panel and 16-channel switch.
- 8. Disconnect the MCR2 tool, and remove the LPCA cover.
- 9. In the LPCA, disconnect the 37-pin MC bias Cable Track cable interfacing to the bulkhead of the failing connector.
- 10. Remove the cable and connector assembly from the MCR2 Tool. Connect the disconnected 37-pin MC bias cable to the 37-pin connector on the MCR2 Tool. Run the TR DD MC diagnostic.
 - a. If the fault is gone then the Connector inside the LPCA appears faulty. Repair or replace the Connector.
 - b. If fault is still present then the fault is further down the chain.
 - c. Reconnect the 37-pin MC bias Cable Track cable to bulkhead in LPCA.
- 11. Swap the two MC bias cables (J18 and J17) at the 16-channel Switch and rerun the diagnostic.
 - a. If fault remains then problem is with Cable Track cable. Check connectors on each end. Either repair issue in connector or replace Cable Track.
 - b. If fault is gone then problem is most likely the 16-channel switch. Inspect the switch, and repair or replace as necessary, refer to the [16 Channel Switch Replacement](#) procedure.
 - c. Return J18 and J17 connections to normal positions.

4.10.3 Forward Production, Fault on Connector A

NOTE: The troubleshooting steps in this section require the MCR3 tool. **Forward Production systems** should only use the MCR3 Tool.

1. **(For Coil connected to A:)** Run MC TR DD Diagnostic with coil connected.
 - a. If failure is seen then, if possible, try running diagnostic with coil connected to Connector **B** and then Connector **C**. If unable to connect coil to **B** or **C** then skip to [Step 2](#).
 - i. If failure still seen when connected to **B** and **C** then check coil connector for damage. Repair or replace the coil.
 - ii. If failures not seen when connected to **B** and **C** then problem is further down MC bias chain on Connector **A** path.
 - b. If failures not seen then recheck error log and confirm problem is really due to MC bias fault on channels between 1 and 8.
2. Remove the coil from Connector **A** and attach the MCR3 Tool. Go to the Multi-Coil Receive 3 Diagnostic menu and under *Port A* select *With Preamps Off*. Select [Run].
 - a. If no failures are seen then this indicates that the receive lines in the connector are in good working order. Skip to [Step 3](#).

- b. If failures are seen then locate the upper row of BNC connectors on the rear of the Multicoil Interface Box and swap the BNC cable on the failing line with one next to it. Run the diagnostic again.
 - i. If the failure moves to another channel then Connector **A** appears faulty. Please repair or replace.
 - ii. If the failure remains on the same channel then the problem may be the preamp. Swap the preamp on the failing channel with one next to it. Keep the cabling in the same location. Run the diagnostic.
 1. If the failure moves to another channel then the preamp is faulty. Refer to the [Multicoil Preamp Replacement](#) procedure.
 2. If the failure remains on the same channel then the problem is further down the chain.
 3. Return the preamp to its original location.
3. Disconnect the cable at J19 on the Multicoil Interface Box and disconnect the 37-pin cable-track cable at the Connector **B** bulkhead interface. Join these two cables together.
 - a. Disconnect all the cables from the upper row of BNC connectors on the rear of the Multicoil Interface Box in the LPCA. Run the MC TR DD Diagnostic.
 - i. If a failure seen then Connector **A** is faulty. Repair or replace.
 - ii. If a failure not seen then the problem is the coil or further down the chain. More tests will determine which.
4. Disconnect J19 from the 37-pin cable-track cable and reconnect 37-pin cable back to normal location. Remove cable at J21 on Multicoil Interface Box and connect this to free J19 cable to bypass Multicoil Interface Box out of the circuit. Run MC TR DD Diagnostic.
 - a. If the fault is gone then the Multicoil Interface Box appears to have failed. Repair or replace the unit.
 - b. If the fault remains then the coil is OK but the problem is further down the chain.
 - c. Reconnect J19 cable to Multicoil Interface Box but leave BNC cables disconnected.
5. Swap cables at J17 and J18 on 1st 16 Channel Switch. Run the MC TR DD Diagnostic.
 - a. If fault seen then replace 37-pin cable-track cable that interconnects J21 on Multicoil Interface Box to J18 on 1st 16 Channel Switch.
 - b. If fault not seen then the problem is further down the chain.
 - c. Reconnect J17 and J18 cables to correct locations. Reconnect BNC cables to correct locations on rear of Multicoil Interface Box.
6. **If system is a 32channel system:** Disconnect connector at J10 on 2nd 16 Channel Switch that services Connector **C**. This 16 Channel Switch is marked with part number 5109645-3 (1.5T) or 5109645-4 (3.0T) and has nothing connected to J21 or J18. Run the MC TR DD Diagnostic.
 - a. If the fault is gone then the problem is between Connector **C** and the 2nd 16 Channel Switch and the coil is OK.
 - b. If the fault remains then the problem is further down the circuit servicing Connector **A**.
7. Open a c-shell window, size it the width of the screen, and in the window type **more /usr/g/service/log/System_Path***. Scroll down to the bottom of the file and find the data with the diagnostic execution time corresponding to the last run of the diagnostic. See [Illustration 14](#).
 - a. Find the fault in Open or Short status column on the right and then match that with a channel on the left. In the Open Ckt Current or Short Ckt Current column note the failing current value.

Illustration 14: System_Path_Diag_Log.txt Output

Diagnostic Execution Time: FEB 10 14:11:37 2008

LPCA Type: Forward Execution System Field Example: 1.1 T

System NOT Configured for H02

If coil has Coil ID coil name shown here

Actual open-circuit and short-circuit currents shown in milliamperes (mA)

MC bias currents, A connector channels

MT Data

Channel	Connector	Short	Obs. Current	Short	Obs. Limit	Open	Obs. Current	Open	Obs. Limit	Short Status	Open Status
1	C9, A1	25	-50	0	0	0	0	No Fault	Not Checked		
2	C10, A1	25	-50	0	0	0	0	No Fault	Not Checked		
3	C11, A1	25	-50	0	0	0	0	No Fault	Not Checked		
4	C12, A1	25	-50	0	0	0	0	No Fault	Not Checked		
5	C13, A1	25	-50	0	0	0	0	No Fault	Not Checked		
6	C14, A1	25	-50	0	0	0	0	No Fault	Not Checked		
7	C15, A1	25	-50	0	0	0	0	No Fault	Not Checked		
8	C16, A1	25	-50	0	0	0	0	No Fault	Not Checked		
9	B1	25	-50	0	0	0	0	No Fault	Not Checked		
10	B2	25	-50	0	0	0	0	No Fault	Not Checked		
11	B3	25	-50	0	0	0	0	No Fault	Not Checked		
12	B4	25	-50	0	0	0	0	No Fault	Not Checked		
13	B5	25	-50	0	0	0	0	No Fault	Not Checked		
14	B6	25	-50	0	0	0	0	No Fault	Not Checked		
15	B7	25	-50	0	0	0	0	No Fault	Not Checked		
16	B8	25	-50	0	0	0	0	No Fault	Not Checked		
17	B9	25	-50	0	0	0	0	No Fault	Not Checked		
18	B10	25	-50	0	0	0	0	No Fault	Not Checked		
19	B11	25	-50	0	0	0	0	No Fault	Not Checked		
20	B12	25	-50	0	0	0	0	No Fault	Not Checked		
21	B13	25	-50	0	0	0	0	No Fault	Not Checked		
22	B14	25	-50	0	0	0	0	No Fault	Not Checked		
23	B15	25	-50	0	0	0	0	No Fault	Not Checked		
24	B16	25	-50	0	0	0	0	No Fault	Not Checked		
25	C1	25	-50	0	0	0	0	No Fault	Not Checked		
26	C2	25	-50	0	0	0	0	No Fault	Not Checked		
27	C3	25	-50	0	0	0	0	No Fault	Not Checked		
28	C4	25	-50	0	0	0	0	No Fault	Not Checked		
29	C5	25	-50	0	0	0	0	No Fault	Not Checked		
30	C6	25	-50	0	0	0	0	No Fault	Not Checked		
31	C7	25	-50	0	0	0	0	No Fault	Not Checked		
32	C8	25	-50	0	0	0	0	No Fault	Not Checked		

8. Swap the connectors at J5 and J10 on the rear of the Driver Module and run the diagnostic again.

- Scroll down the c-shell window to find the most recent data.
- Swapping the cables means that the MC bias lines for the coil have shifted up 16 places. This means that if the failure was on channel 5 before it could now be on channel 21. Ignore all the new failures now reported in the Open Status and Short Status columns, as well as any failures seen on the Diagnostic GUI.
 - If the failing current value is now present on one of the channels between 17 and 24 then the problem is between the 1st 16 Channel Switch and the cabling to the Driver Module. Driver Module is OK.
 - If the failing current value is not seen between 17 and 24 then the problem is inside the Driver Module. Repair or replace the Driver Module, refer to the [Driver Module Replacement](#) procedure.
- Swap connectors at J5 and J10 on rear of Driver Module back to normal positions.

9. Swap connectors at J78 and J203 on Equipment Room-side of Penetration Panel and run the diagnostic again.

- Scroll down the c-shell window to find the most recent data. Again, ignore failure messages in far right column.
- Verify that the same channel seen in the previous steps is still reporting an abnormal current value.
 - If the failure is still present on same channel between 17 and 24 then problem is between Pen Panel filters and 16 Channel Switch.
 - If the same channel between 17 and 24 that was failing before is passing now then replace or repair cable between J5 on Driver Module and Penetration Panel.
 - Swap connectors at J78 and J203 on Equipment Room-side of Penetration Panel back to normal positions.

10. Swap connectors at J78 and J203 on Magnet Room-side of Penetration Panel and run the diagnostic again.

- Scroll down the c-shell window to find the most recent data. Again, ignore failure messages in far right column.
- Verify that the same channel seen in the previous steps is still reporting an abnormal current value.
 - If the failure is still present on same channel between 17 and 24 then problem is between the Magnet Room MC bias cabling and 16 Channel Switch.
 - If the same channel between 17 and 24 that was failing before is passing now then replace the failed Penetration Panel filter at J203.

- iii. Swap connectors at J78 and J203 on Magnet Room-side of Penetration Panel back to normal positions.
- c. Swap connectors at J10 and J11 on 1st 16 Channel Switch and run the diagnostic again.
- d. Scroll down the c-shell window to find the most recent data. Again, ignore failure messages in far right column.
- e. Verify that the same channel seen in the previous steps is still reporting an abnormal current value.
 - i. If the failure is still present on same channel between 17 and 24 then problem is the 16 Channel Switch. Replace or repair 16 Channel Switch, refer to the [16 Channel Switch Replacement](#) procedure.
 - ii. If the same channel between 17 and 24 that was failing before is passing now then repair or replace the MC bias cable connected to J203 on the Magnet Room-side of the Penetration Panel.
 - iii. Swap connectors at J10 and J11 on 1st 16 Channel Switch back to normal positions.

4.10.4 Forward Production, Fault on Connector B, Coil Ch 1-8 (Bias Ch 9-16) ONLY

NOTE: The troubleshooting steps in this section require the MCR3 tool. **Forward Production** systems should only use the MCR3 Tool.

1. **Coil connected to B:** Run MC TR DD Diagnostic with coil connected.
 - a. If failure is seen then, if possible, try running diagnostic with coil connected to Connector **A** and then Connector **C**. If unable to connect coil to **A** or **C** then skip to [Step 2](#).
 - i. If failure still seen when connected to **A** and **C** then check coil connector for damage. Repair or replace the coil.
 - ii. If failures not seen when connected to **A** and **C** then problem is further down MC bias chain on Connector **B** path.
 - b. If failures not seen then recheck error log and confirm problem is really due to MC bias fault on channels between 9 and 16.
2. Remove the coil from Connector **B** and attach the MCR3 Tool. Go to the Multi-Coil Receive 3 Diagnostic menu and under *Port B* select *Hypertronics* and [Run].
 - a. If no failures are seen then this indicates that the receive lines in the connector are in good working order. Skip to [Step 3](#).
 - b. If failures are seen then remove the LPCA cover. Disconnect the 37-pin connector and upper 8W8 RF connector in the Connector **B** bulkhead interface.
 - c. Remove the cable assembly from the MCR3 Tool and connect the 37-pin connector and 8W8 RF connector directly to the MCR3 Tool. Run the diagnostic again.
 - i. If failure not seen then Connector **B** faulty. Please repair or replace.
 - ii. If failure remains then disconnect cables at J1 and J5 on Image Reject Filter 1. Bypass Filter by connecting cables together. Run the Multi-coil Receive 3 diagnostic.
 1. If failure not seen then Image Reject Filter 1 is faulty. Please repair or replace.
 2. If failure remains then repair or replace 8W8 RF cable-track cable.
 - d. Disconnect the 8W8 RF cable from the MCR3 Tool but leave the 37-pin cable connected. A connection to 8W8 RF connector on MCR3 Tool not needed. Run the MC TR DD Diagnostic.
 - i. If failure not seen then Connector **B** is faulty. Repair or replace.
 - ii. If failure seen then the problem is further down the chain.
3. Disconnect J2 and J6 from the Image Reject Filter and bypass it from the circuit by connected these two cables together. Run MC TR DD Diagnostic.

- a. If fault is gone then the Image Reject Filter is faulty. Repair or replace.
 - b. If fault remains then the problem is further down the chain.
4. Disconnect J6 from J2 and attach MCR3 Tool to J6 37-pin connector. A connection to 8W8 RF connector on MCR3 Tool not needed. Run *MC TR DD Diagnostic*.
 - a. If fault gone the 37-pin cable in cable track is faulty. Repair or replace cable-track.
 - b. If fault remains then problem is further down the chain.
 - c. Return J6 and J2 to their proper positions on the Image Reject Filter Box.
 5. Swap connectors at J17 and J18 on 1st 16 Channel Switch. Run the diagnostic.
 - a. If fault gone then problem is between 1st 16 Channel Switch 1 and Driver Module.
 - b. If fault remains then repair or replace cable that interconnects J6 on Image Reject Filter 1 to J17 on 1st 16 Channel Switch.
 - c. Reconnect J17 and J18 cables to correct locations. Run diagnostic.
 6. Open a c-shell window, size it the width of the screen, and in the window type **more /usr/g/service/log/System_Path***. Scroll down to the bottom of the file and find the data with the diagnostic execution time corresponding to the last run of the diagnostic. See [Illustration 15](#).

Illustration 15: System_Path_Diag_Log.txt Output

Diagnostic Execution Time: FEB 18 14:53:27 2008

LPFA Type: Forward Production System Field Strength: 1.5 T

System NOT Configured for MRI

Connector Multicoinc 1: No Coil Detected
 Connector Multicoinc 1A: No Coil Detected
 Connector Multicoinc 2: No Coil Detected

MC Data

If coil has Coil ID coil name shown here

Actual open circuit and short circuit currents shown in milliamperes (mA)

Channel	Connector	Short Ckt Current	Short Ckt Limit	Open Ckt Current	Open Ckt Limit	Short Status	Open Status
1	CS - A1	0	-50	0	0	No Fault	Not Checked
2	CL1 - A1	0	-50	0	0	No Fault	Not Checked
3	CL2 - A1	0	-50	0	0	No Fault	Not Checked
4	CL3 - A1	0	-50	0	0	No Fault	Not Checked
5	CL4 - A1	0	-50	0	0	No Fault	Not Checked
6	CL5 - A1	0	-50	0	0	No Fault	Not Checked
7	CL6 - A1	0	-50	0	0	No Fault	Not Checked
8	CL7 - A1	0	-50	0	0	No Fault	Not Checked
9	CL8 - A1	0	-50	0	0	No Fault	Not Checked
10	CL9 - A1	0	-50	0	0	No Fault	Not Checked
11	CL10 - A1	0	-50	0	0	No Fault	Not Checked
12	CL11 - A1	0	-50	0	0	No Fault	Not Checked
13	CL12 - A1	0	-50	0	0	No Fault	Not Checked
14	CL13 - A1	0	-50	0	0	No Fault	Not Checked
15	CL14 - A1	0	-50	0	0	No Fault	Not Checked
16	CL15 - A1	0	-50	0	0	No Fault	Not Checked
17	CL16 - A1	0	-50	0	0	No Fault	Not Checked
18	CL17 - A1	0	-50	0	0	No Fault	Not Checked
19	CL18 - A1	0	-50	0	0	No Fault	Not Checked
20	CL19 - A1	0	-50	0	0	No Fault	Not Checked
21	CL20 - A1	0	-50	0	0	No Fault	Not Checked
22	CL21 - A1	0	-50	0	0	No Fault	Not Checked
23	CL22 - A1	0	-50	0	0	No Fault	Not Checked
24	CL23 - A1	0	-50	0	0	No Fault	Not Checked
25	CL24 - A1	0	-50	0	0	No Fault	Not Checked
26	CL25 - A1	0	-50	0	0	No Fault	Not Checked
27	CL26 - A1	0	-50	0	0	No Fault	Not Checked
28	CL27 - A1	0	-50	0	0	No Fault	Not Checked
29	CL28 - A1	0	-50	0	0	No Fault	Not Checked
30	CL29 - A1	0	-50	0	0	No Fault	Not Checked
31	CL30 - A1	0	-50	0	0	No Fault	Not Checked
32	CL31 - A1	0	-50	0	0	No Fault	Not Checked
33	CL32 - A1	0	-50	0	0	No Fault	Not Checked
34	CL33 - A1	0	-50	0	0	No Fault	Not Checked
35	CL34 - A1	0	-50	0	0	No Fault	Not Checked
36	CL35 - A1	0	-50	0	0	No Fault	Not Checked
37	CL36 - A1	0	-50	0	0	No Fault	Not Checked
38	CL37 - A1	0	-50	0	0	No Fault	Not Checked
39	CL38 - A1	0	-50	0	0	No Fault	Not Checked
40	CL39 - A1	0	-50	0	0	No Fault	Not Checked
41	CL40 - A1	0	-50	0	0	No Fault	Not Checked
42	CL41 - A1	0	-50	0	0	No Fault	Not Checked
43	CL42 - A1	0	-50	0	0	No Fault	Not Checked
44	CL43 - A1	0	-50	0	0	No Fault	Not Checked
45	CL44 - A1	0	-50	0	0	No Fault	Not Checked
46	CL45 - A1	0	-50	0	0	No Fault	Not Checked
47	CL46 - A1	0	-50	0	0	No Fault	Not Checked
48	CL47 - A1	0	-50	0	0	No Fault	Not Checked
49	CL48 - A1	0	-50	0	0	No Fault	Not Checked
50	CL49 - A1	0	-50	0	0	No Fault	Not Checked
51	CL50 - A1	0	-50	0	0	No Fault	Not Checked
52	CL51 - A1	0	-50	0	0	No Fault	Not Checked
53	CL52 - A1	0	-50	0	0	No Fault	Not Checked
54	CL53 - A1	0	-50	0	0	No Fault	Not Checked
55	CL54 - A1	0	-50	0	0	No Fault	Not Checked
56	CL55 - A1	0	-50	0	0	No Fault	Not Checked
57	CL56 - A1	0	-50	0	0	No Fault	Not Checked
58	CL57 - A1	0	-50	0	0	No Fault	Not Checked
59	CL58 - A1	0	-50	0	0	No Fault	Not Checked
60	CL59 - A1	0	-50	0	0	No Fault	Not Checked
61	CL60 - A1	0	-50	0	0	No Fault	Not Checked
62	CL61 - A1	0	-50	0	0	No Fault	Not Checked
63	CL62 - A1	0	-50	0	0	No Fault	Not Checked
64	CL63 - A1	0	-50	0	0	No Fault	Not Checked
65	CL64 - A1	0	-50	0	0	No Fault	Not Checked
66	CL65 - A1	0	-50	0	0	No Fault	Not Checked
67	CL66 - A1	0	-50	0	0	No Fault	Not Checked
68	CL67 - A1	0	-50	0	0	No Fault	Not Checked
69	CL68 - A1	0	-50	0	0	No Fault	Not Checked
70	CL69 - A1	0	-50	0	0	No Fault	Not Checked
71	CL70 - A1	0	-50	0	0	No Fault	Not Checked
72	CL71 - A1	0	-50	0	0	No Fault	Not Checked
73	CL72 - A1	0	-50	0	0	No Fault	Not Checked
74	CL73 - A1	0	-50	0	0	No Fault	Not Checked
75	CL74 - A1	0	-50	0	0	No Fault	Not Checked
76	CL75 - A1	0	-50	0	0	No Fault	Not Checked
77	CL76 - A1	0	-50	0	0	No Fault	Not Checked
78	CL77 - A1	0	-50	0	0	No Fault	Not Checked
79	CL78 - A1	0	-50	0	0	No Fault	Not Checked
80	CL79 - A1	0	-50	0	0	No Fault	Not Checked
81	CL80 - A1	0	-50	0	0	No Fault	Not Checked
82	CL81 - A1	0	-50	0	0	No Fault	Not Checked
83	CL82 - A1	0	-50	0	0	No Fault	Not Checked
84	CL83 - A1	0	-50	0	0	No Fault	Not Checked
85	CL84 - A1	0	-50	0	0	No Fault	Not Checked
86	CL85 - A1	0	-50	0	0	No Fault	Not Checked
87	CL86 - A1	0	-50	0	0	No Fault	Not Checked
88	CL87 - A1	0	-50	0	0	No Fault	Not Checked
89	CL88 - A1	0	-50	0	0	No Fault	Not Checked
90	CL89 - A1	0	-50	0	0	No Fault	Not Checked
91	CL90 - A1	0	-50	0	0	No Fault	Not Checked
92	CL91 - A1	0	-50	0	0	No Fault	Not Checked
93	CL92 - A1	0	-50	0	0	No Fault	Not Checked
94	CL93 - A1	0	-50	0	0	No Fault	Not Checked
95	CL94 - A1	0	-50	0	0	No Fault	Not Checked
96	CL95 - A1	0	-50	0	0	No Fault	Not Checked
97	CL96 - A1	0	-50	0	0	No Fault	Not Checked
98	CL97 - A1	0	-50	0	0	No Fault	Not Checked
99	CL98 - A1	0	-50	0	0	No Fault	Not Checked
100	CL99 - A1	0	-50	0	0	No Fault	Not Checked
101	CL100 - A1	0	-50	0	0	No Fault	Not Checked
102	CL101 - A1	0	-50	0	0	No Fault	Not Checked
103	CL102 - A1	0	-50	0	0	No Fault	Not Checked
104	CL103 - A1	0	-50	0	0	No Fault	Not Checked
105	CL104 - A1	0	-50	0	0	No Fault	Not Checked
106	CL105 - A1	0	-50	0	0	No Fault	Not Checked
107	CL106 - A1	0	-50	0	0	No Fault	Not Checked
108	CL107 - A1	0	-50	0	0	No Fault	Not Checked
109	CL108 - A1	0	-50	0	0	No Fault	Not Checked
110	CL109 - A1	0	-50	0	0	No Fault	Not Checked
111	CL110 - A1	0	-50	0	0	No Fault	Not Checked
112	CL111 - A1	0	-50	0	0	No Fault	Not Checked
113	CL112 - A1	0	-50	0	0	No Fault	Not Checked
114	CL113 - A1	0	-50	0	0	No Fault	Not Checked
115	CL114 - A1	0	-50	0	0	No Fault	Not Checked
116	CL115 - A1	0	-50	0	0	No Fault	Not Checked
117	CL116 - A1	0	-50	0	0	No Fault	Not Checked
118	CL117 - A1	0	-50	0	0	No Fault	Not Checked
119	CL118 - A1	0	-50	0	0	No Fault	Not Checked
120	CL119 - A1	0	-50	0	0	No Fault	Not Checked
121	CL120 - A1	0	-50	0	0	No Fault	Not Checked
122	CL121 - A1	0	-50	0	0	No Fault	Not Checked
123	CL122 - A1	0	-50	0	0	No Fault	Not Checked
124	CL123 - A1	0	-50	0	0	No Fault	Not Checked
125	CL124 - A1	0	-50	0	0	No Fault	Not Checked
126	CL125 - A1	0	-50	0	0	No Fault	Not Checked
127	CL126 - A1	0	-50	0	0	No Fault	Not Checked
128	CL127 - A1	0	-50	0	0	No Fault	Not Checked
129	CL128 - A1	0	-50	0	0	No Fault	Not Checked
130	CL129 - A1	0	-50	0	0	No Fault	Not Checked
131	CL130 - A1	0	-50	0	0	No Fault	Not Checked
132	CL131 - A1	0	-50	0	0	No Fault	Not Checked
133	CL132 - A1	0	-50	0	0	No Fault	Not Checked
134	CL133 - A1	0	-50	0	0	No Fault	Not Checked
135	CL134 - A1	0	-50	0	0	No Fault	Not Checked
136	CL135 - A1	0	-50	0	0	No Fault	Not Checked
137	CL136 - A1	0	-50	0	0	No Fault	Not Checked
138	CL137 - A1	0	-50	0	0	No Fault	Not Checked
139	CL138 - A1	0	-50	0	0	No Fault	Not Checked
140	CL139 - A1	0	-50	0	0	No Fault	Not Checked
141	CL140 - A1	0	-50	0	0	No Fault	Not Checked
142	CL141 - A1	0	-50	0	0	No Fault	Not Checked
143	CL142 - A1	0	-50	0	0	No Fault	Not Checked
144	CL143 - A1	0	-50	0	0	No Fault	Not Checked
145	CL144 - A1	0	-50	0	0	No Fault	Not Checked
146	CL145 - A1	0	-50	0	0	No Fault	Not Checked
147	CL146 - A1	0	-50	0	0	No Fault	Not Checked
148	CL147 - A1	0	-50	0	0	No Fault	Not Checked
149	CL148 - A1	0	-50	0	0	No Fault	Not Checked
150	CL149 - A1	0	-50	0	0	No Fault	Not Checked
151	CL150 - A1	0	-50	0	0	No Fault	Not Checked
152	CL151 - A1	0	-50	0	0	No Fault	Not Checked
153	CL152 - A1	0	-50	0	0	No Fault	Not Checked
154	CL153 - A1	0	-50	0	0	No Fault	Not Checked
155	CL154 - A1	0	-50	0	0	No Fault	Not Checked
156	CL155 - A1	0	-50	0	0	No Fault	Not Checked
157	CL156 - A1	0	-50	0	0	No Fault	Not Checked
158	CL157 - A1	0	-50	0	0	No Fault	Not Checked
159	CL158 - A1	0	-50	0	0	No Fault	Not Checked
160	CL159 - A1	0	-50	0	0	No Fault	Not Checked
161	CL160 - A1	0	-50	0	0	No Fault	Not Checked
162	CL161 - A1	0	-50	0	0	No Fault	Not Checked
163	CL162 - A1	0	-50	0	0	No Fault	Not Checked
164	CL163 - A1	0	-50	0	0	No Fault	Not Checked
165	CL164 - A1	0	-50	0	0	No Fault	Not Checked
166	CL165 - A1	0	-50	0	0	No Fault	Not Checked
167	CL166 - A1	0	-50	0	0	No Fault	Not Checked
168	CL167 - A1	0	-50	0	0	No Fault	Not Checked
169	CL168 - A1	0	-50	0	0	No Fault	Not Checked
170	CL169 - A1	0	-50	0	0	No Fault	Not Checked
171	CL170 - A1	0	-50	0	0	No Fault	Not Checked
172	CL171 - A1	0	-50	0	0	No Fault	Not Checked
173	CL172 - A1	0	-50	0	0	No Fault	Not Checked
174	CL173 - A1	0	-50	0	0	No Fault	Not Checked
175	CL174 - A1	0	-50	0	0	No Fault	Not Checked
176	CL175 - A1	0	-50	0	0	No Fault	Not Checked
177	CL176 - A1	0	-50	0	0	No Fault	Not Checked
178	CL177 - A1	0	-50	0	0	No Fault	Not Checked
179	CL178 - A1	0	-50	0	0	No Fault	Not Checked
180	CL179 - A1	0	-50	0	0	No Fault	Not Checked
181	CL180 - A1	0	-50	0	0	No Fault	Not Checked
182	CL181 - A1	0	-50	0	0	No Fault	Not Checked
183	CL182 - A1	0	-50	0	0	No Fault	Not Checked
184	CL183 - A1	0	-50	0	0	No Fault	Not Checked
185	CL184 - A1	0	-50	0	0	No Fault	Not Checked
186	CL185 - A1	0	-50	0	0	No Fault	Not Checked
187	CL186 - A1						

- ii. If failing current value is not seen in channels between 25 and 32 then the problem is inside the Driver Module. Repair or replace the Driver Module, refer to the [Driver Module Replacement](#) procedure.
- d. Swap connectors at J5 and J10 on rear of Driver Module back to normal positions.
- 8. Swap connectors at J78 and J203 on Equipment Room-side of Penetration Panel and run the diagnostic again.
 - a. Scroll down the c-shell window to find the most recent data. Again, ignore failure messages in far right column.
 - b. Verify that the same channel seen in the previous steps is still reporting an abnormal current value.
 - i. If the failure is still present on same channel between 25 and 32 then problem is between Pen Panel filters and 16 Channel Switch.
 - ii. If the same channel between 25 and 32 that was failing before is passing now then replace or repair cable between J5 on Driver Module and Penetration Panel.
 - c. Swap connectors at J78 and J203 on Equipment Room-side of Penetration Panel back to normal positions.
- 9. Swap connectors at J78 and J203 on Magnet Room-side of Penetration Panel and run the diagnostic again.
 - a. Scroll down the c-shell window to find the most recent data. Again, ignore failure messages in far right column.
 - b. Verify that the same channel seen in the previous steps is still reporting an abnormal current value.
 - i. If the failure is still present on same channel between 25 and 32 then problem is between the Magnet Room MC bias cabling and 16 Channel Switch.
 - ii. If the same channel between 25 and 32 that was failing before is passing now then replace the failed Penetration Panel filter at J203.
 - iii. Swap connectors at J78 and J203 on Magnet Room-side of Penetration Panel back to normal positions.
- 10. Swap connectors at J10 and J11 on 1st 16 Channel Switch and run the diagnostic again.
 - a. Scroll down the c-shell window to find the most recent data. Again, ignore failure messages in far right column.
 - b. Verify that the same channel seen in the previous steps is still reporting an abnormal current value.
 - i. If the failure is still present on same channel between 25 and 32 then problem is the 16 Channel Switch. Replace or repair 16 Channel Switch.
 - ii. If the same channel between 25 and 32 that was failing before is passing now then repair or replace the MC bias cable connected to J203 on the Magnet Room-side of the Penetration Panel.
 - c. Swap connectors at J10 and J11 on 1st 16 Channel Switch back to normal positions.

4.10.5 Forward Production, Fault on Connector B, Coil Ch 9-16 (Bias Ch 17-24) ONLY

NOTE: The troubleshooting steps in this section require the MCR3 tool. **Forward Production systems** should only use the MCR3 Tool.

- 1. Run MC TR DD Diagnostic with coil connected.
 - a. If failures seen then, if possible, try running diagnostic with coil connected to Connector **C**. If no Connector **C** then skip to [Step 2](#).
 - i. If failures still seen when connected to **C** then check coil connector for damage. Repair or replace coil.
 - ii. If failures not seen when connected to **C** then problem is further down MC bias chain on Connector **B** path.

- b. If failures not seen then recheck error log and confirm problem is really due to MC bias fault on channels between 17 and 24.
2. Remove the coil from Connector **B** and remove the LPCA cover. Swap upper and lower 8W8 RF cables connected to Connector **B** bulkhead interface. Attach the MCR3 Tool to Connector **B**.
3. Go to the Multi-Coil Receive 3 Diagnostic menu and under *Port B* select *Upper 8W8* and [Run].
 - a. If no failures are seen then this indicates that the connector and receive lines are in good working order. Skip to [Step 4](#).
 - b. If failures seen then remove 37-pin cable and bottom-most 8W8 RF connector from Connector **B** bulkhead interface. Also remove 37-pin connector from same bulkhead.
 - c. Remove the cable assembly from the MCR3 Tool and connect the 37-pin connector and 8W8 RF connector directly to the MCR3 Tool. Run the Multi-Coil Receive 3 diagnostic again.
 - i. If failure not seen then Connector **B** is faulty. Please repair or replace.
 - ii. If failure remains then disconnect cables at J3 and J7 on Image Reject Filter 1. Bypass Filter by connecting cables together. Run the MCR diagnostic.
 - iii. If failure not seen then Image Reject Filter 1 is faulty. Please repair or replace.
 - iv. If failure remains then repair or replace 8W8 RF cable-track cable.
4. Disconnect the 37-pin cable but leave the 8W8 RF cable connected. A connection to 37-pin cable on MCR3 Tool not needed.
 - a. Run the MC TR DD Diagnostic. It will report that no coil is connected. This is normal.
5. Open a c-shell window, size it the width of the screen, and in the window type **more /usr/g/service/log/System_Path***. Scroll down to the bottom of the file and find the data with the diagnostic execution time corresponding to the last run of the diagnostic. See [Illustration 16](#).
 - a. Find the fault in Open or Short status column on the right and then match that with a channel on the left. In the Open Ckt Current or Short Ckt Current column note the failing current value.

Illustration 16: System_Path_Diag_Log.txt output

Diagnostic Execution Time: FEB 18 18:57:37 1998

LPCA Type: Forward Production System Field Strength: 1.5 T

System NOT configured for HSE

Connector Multicoil C: No Coil Detected
 Connector Multicoil A: No Coil Detected
 Connector Multicoil B: No Coil Detected

FE Data

Actual open - circuit and short - circuit currents shown in milliamperes (mA)

Channel	Connector	Short Ckt Current	Short Ckt Limit	Open Ckt Current	Open Ckt Limit	Short Status	Open Status
1	CL1 - A1	0	-50	0	0	No Fault	Not Checked
2	CL2 - A1	0	-50	0	0	No Fault	Not Checked
3	CL3 - A1	0	-50	0	0	No Fault	Not Checked
4	CL4 - A1	0	-50	0	0	No Fault	Not Checked
5	CL5 - A1	0	-50	0	0	No Fault	Not Checked
6	CL6 - A1	0	-50	0	0	No Fault	Not Checked
7	CL7 - A1	0	-50	0	0	No Fault	Not Checked
8	CL8 - A1	0	-50	0	0	No Fault	Not Checked
9	CL9 - A1	0	-50	0	0	No Fault	Not Checked
10	CL10 - A1	0	-50	0	0	No Fault	Not Checked
11	CL11 - A1	0	-50	0	0	No Fault	Not Checked
12	CL12 - A1	0	-50	0	0	No Fault	Not Checked
13	CL13 - A1	0	-50	0	0	No Fault	Not Checked
14	CL14 - A1	0	-50	0	0	No Fault	Not Checked
15	CL15 - A1	0	-50	0	0	No Fault	Not Checked
16	CL16 - A1	0	-50	0	0	No Fault	Not Checked
17	CL17 - A1	-12	-50	-5	0	No Fault	Not Checked
18	CL18 - A1	-15	-50	-5	0	No Fault	Not Checked
19	CL19 - A1	-15	-50	-10	0	No Fault	Not Checked
20	CL20 - A1	-5	-50	-10	0	No Fault	Not Checked
21	CL21 - A1	-10	-50	-5	0	No Fault	Not Checked
22	CL22 - A1	0	-50	-5	0	No Fault	Not Checked
23	CL23 - A1	-5	-50	-5	0	No Fault	Not Checked
24	CL24 - A1	0	-50	0	0	No Fault	Not Checked
25	CL25 - A1	0	-50	0	0	No Fault	Not Checked
26	CL26 - A1	0	-50	0	0	No Fault	Not Checked
27	CL27 - A1	0	-50	0	0	No Fault	Not Checked
28	CL28 - A1	0	-50	0	0	No Fault	Not Checked
29	CL29 - A1	0	-50	0	0	No Fault	Not Checked
30	CL30 - A1	0	-50	0	0	No Fault	Not Checked
31	CL31 - A1	0	-50	0	0	No Fault	Not Checked
32	CL32 - A1	0	-50	0	0	No Fault	Not Checked

MC bias currents for B connector channels

6. Swap the connectors at J5 and J10 on the rear of the Driver Module and run the diagnostic again.
 - a. Scroll down the c-shell window to find the most recent data.

- b. Swapping the cables means that the MC bias lines for the coil have shifted down 16 places. This means that if the failure was on channel 17 before it could now be on channel 1 and if it was on channel 23 before it could now be on channel 7. Ignore all the new failures now reported in the Open Status and Short Status columns.
 - i. If the failing current value is now present on one of the channels between 1 and 8 then the problem is between the 1st 16 Channel Switch and the cabling to the Driver Module. Driver Module is OK.
 - ii. If failing current value is not seen in channels between 1 and 8 then the problem is inside the Driver Module. Repair or replace the Driver Module, refer to the [Driver Module Replacement](#) procedure.
- c. Swap connectors at J5 and J10 on rear of Driver Module back to normal positions.
- 7. Swap connectors at J78 and J203 on Equipment Room-side of Penetration Panel and run the diagnostic again.
 - a. Scroll down the c-shell window to find the most recent data. Again, ignore failure messages in far right column.
 - b. Verify that the same channel seen in the previous steps is still reporting an abnormal current value.
 - i. If the failure is still present on same channel between 1 and 8 then problem is between Pen Panel filters and 16 Channel Switch.
 - ii. If the same channel between 1 and 8 that was failing before is passing now then replace or repair cable between J10 on Driver Module and Penetration Panel.
 - c. Swap connectors at J78 and J203 on Equipment Room-side of Penetration Panel back to normal positions.
- 8. Swap connectors at J78 and J203 on Magnet Room-side of Penetration Panel and run the diagnostic again.
 - a. Scroll down the c-shell window to find the most recent data. Again, ignore failure messages in far right column.
 - b. Verify that the same channel seen in the previous steps is still reporting an abnormal current value.
 - i. If the failure is still present on same channel between 1 and 8 then problem is between the Magnet Room MC bias cabling and 16 Channel Switch.
 - ii. If the same channel between 1 and 8 that was failing before is passing now then replace the failed Penetration Panel filter at J78.
 - c. Swap connectors at J78 and J203 on Magnet Room-side of Penetration Panel back to normal positions.
- 9. Swap connectors at J10 and J11 on 1st 16 Channel Switch and run the diagnostic again.
 - a. Scroll down the c-shell window to find the most recent data. Again, ignore failure messages in far right column.
 - b. Verify that the same channel seen in the previous steps is still reporting an abnormal current value.
 - i. If the failure is still present on same channel between 1 and 8 then problem is either the 1st 16 Channel Switch or cable connecting to J19 on the 1st 16 Channel Switch. The cable is identical in construction to a standard 25-pin serial cable and pin connections are one-to-one from one end of the cable to the other. Replace or repair the cable or 16 Channel Switch, refer to the [16 Channel Switch Replacement](#) procedure.
 - ii. If the same channel between 1 and 8 that was failing before is passing now then repair or replace the MC bias cable connected to J78 on the Magnet Room-side of the Penetration Panel.
 - iii. Swap connectors at J10 and J11 on 1st 16 Channel Switch back to normal positions.

4.10.6 Forward Production, Fault on Connector C, Ch.25-32 ONLY

- 1. Content in development.

4.10.7 Forward Production, Fault on Connector C, Ch.1-8 ONLY

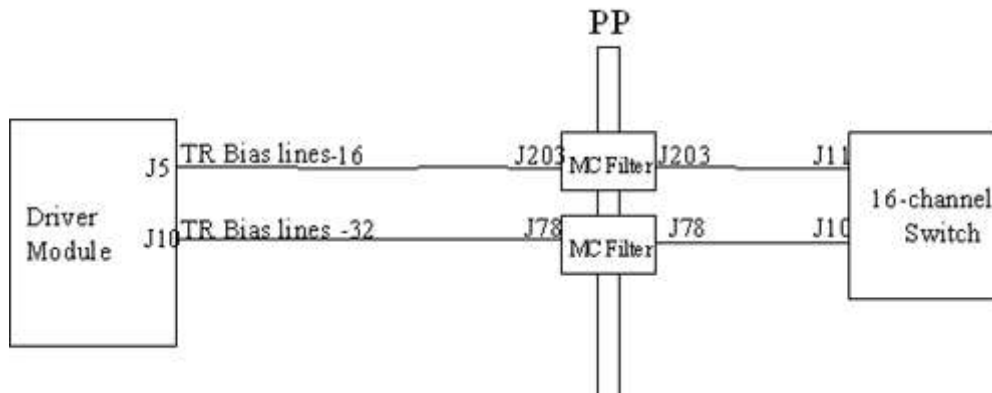
1. Content in development.

4.11 MC Short Circuit Fault

If the system is reporting a MC Short Circuit Fault, troubleshooting must occur to isolate it to the failed component. [Illustration 17](#) shows a path for the MC signals.

1. Un-plug the coil and repeat the Diagnostic, to verify it is not a short in the coil.
 - a. If still get the error, then short is in the system path, and is **NOT** the coil
 - b. If no longer getting the short circuit fault, the issue is within the coil.

Illustration 17: System MC Circuit Path



Start disconnecting cables along the path to determine where the short is located.

2. Disconnect the MC cable containing the channel in question at Driver Module (J5 or J10) and repeat the Diagnostic.
 - a. If still get a short, then issue is in the Driver Module. Replace the Driver Module, refer to the [Driver Module Replacement](#) procedure.
 - b. If only get an open circuit fault (due to the disconnected cable), then short is further along the path past the induced open, and is **NOT** the Driver Module. Continue to troubleshoot.
 - c. Reconnect the cable at the driver module
3. Disconnect the MC cable at the 16-channel Switch (J10 or J11) and repeat the Diagnostic.
 - a. If still get a short, then issue is in the cable chain between Driver Module and 16-channel switch. Continue troubleshooting at [Step 4](#).
 - b. If the short goes away, and only an Open circuit fault is detected, then problem is most likely within the 16-channel switch, LPCA circuitry, or Coil.
 - i. Remove the connector from J19 on the 16- Channel Switch and run the diagnostic.
 1. If the failure is gone then the problem is further down the chain.
 2. If the failure remains then replace the 16 Channel Switch, refer to the [16 Channel Switch Replacement](#) procedure.
 - ii. Remove the connector from J9 on the Multicoil Integrated Assembly box in the LPCA. Run the diagnostic.
 1. If the failure is gone then it is possible that the preamp on that receive line is faulty.
 - a. Inside the Multicoil Preamp Assembly swap the preamp on the failing channel with one next to it. Run the diagnostic.
 - b. If the fault remains on the same channel then replace the Multicoil Integrated Assembly box, refer to the [Multicoil Preamp Replacement](#) procedure.

c. If the fault moves to another channel then replace the failed preamp.

2. If the failure remains then replace or repair the cable-track cable.

c. Reconnect the cable at the 16-channel Switch.

4. Disconnect MC cable at Equipment room side of the Pen-panel (J78 or J203) and repeat the Diagnostic.

a. If still get a short, then issue is in the cable between Driver Module and Pen-panel. Inspect the cable and connectors, and repair or replace the cable.

b. If the short goes away, and only get an Open circuit fault, then problem is most likely the MC Filter or cable in the Magnet room Reconnect the cable. Continue with [Step 5](#).

5. Disconnect the MC cable (J203 or J78) on the Magnet room side of the Pen Panel, and rerun the diagnostic.

a. If still continue to see a short on the original MC line, then the issue is the MC filter on the Pen Panel. Replace the filter.

b. If the short goes away, and only get an open circuit fault, repair or replace the cable between the Pen Panel and the 16-channel switch.

c. Return the disconnected cable to the appropriate position.

4.12 MC Receive Fault

If the diagnostic is detecting a MC Receive Fault, it is detecting current of greater than +100 mA during receive mode. The most likely cause of this failure is a MC Driver in the Driver Module. Replace the Driver Module, refer to the [Driver Module Replacement](#) procedure.

NOTE: In rare instances this error has been received due to instability in the MC bias path caused by a capacitor in the circuit. If this error only occurs on one coil, there is a small chance it could be the coil circuitry causing the fault.

4.13 MC Transmit Fault

If the diagnostic is detecting a MC Transmit Fault, it is detecting negative current of less than -50 mA during transmit mode. The most likely cause of this failure is a Drivers or Monitoring Circuit within the Driver Module. Replace the Driver Module, refer to the [Driver Module Replacement](#) procedure.

NOTE: In rare instances this error has been received due to instability in the MC bias path caused by a capacitor in the circuit. If this error only occurs on one coil, there is a small chance it could be the coil circuitry causing the fault.

4.14 Head and Body TR Cable Swap

The system should be able to detect if the Body and Head T/R cables have been swapped, based on the difference in voltages. If this error is received, check the cabling at both the output of the Driver Module (J6 and J7), as well as at the input of the RF Amplifier (J9 and J10). At the output of the Amplifier, the cables are no longer the same cable connector, so they will not be swapped anywhere else along the chain.

4.15 Head and Connector A Cable Swap

Upgrade systems: the Head T/R signal is split into two paths at the RF TX Switch on the Pen-panel. If this error is received, check the cabling at both the Output of the RF TX Switch (J151 and J153), as well as at the Cable I/F Panel at the Rear Pedestal (J1 and J2).

4.16 MC Cable Swap

Upgrade systems: the Diagnostic can detect potential MC Cable swaps along the MC Paths. If this error is received, check the cabling at the back of the Driver Module (J5 and J10), cables at the Pen Panel (J203 and J78), and cables at the 16-Channel Switch (J11 and J10).

4.17 MC Cable Disconnected

The Diagnostic can detect when an MC Cable is disconnected along the MC Paths. If this error is received, check the cabling at the back of the Driver Module (J5 and J10), cables at the Pen Panel (J203 and J78), and cables at the 16-Channel Switch (J11 and J10).

5 Finalization

1. Return all equipment to the original configuration. Be sure to connect all cables, and verify swapped cables were returned to correct locations according to the cable maps.
2. Remove all LOTO components and turn on the RF Amp breakers, if turned off.
3. Run the TR DD MC System Path Diag a final time to verify all cables are returned properly, and no new faults have appeared.
4. Perform a TPS Reset.
5. Complete one Head, Body, and MNS (if applicable) scan to verify system functionality.

 [Back to top](#)